

SAGE

FRAMEWORK

Structured AI-Guided Education

Embedding Generative AI in Curriculum Evidence-Based Implementation Guide

A Research-Informed, Ready-to-Adopt Method for Unit Coordinators and Course Designers
Supporting Academic Integrity, Assessment Assurance, and AI Literacy Development



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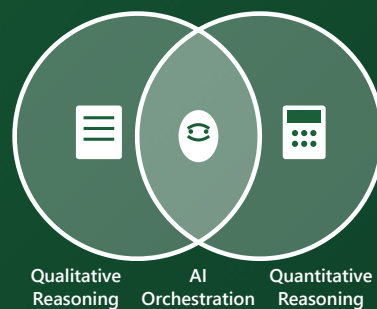
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AI Literacy and Numeracy: A Unified Framework

This resource adopts the paired terms "**AI literacy and numeracy**" to recognize the full spectrum of GenAI capabilities in higher education. AI literacy encompasses critical evaluation, ethical reasoning, and qualitative analysis skills required for text-based AI interactions. AI numeracy extends this to quantitative domains including data analysis, mathematical problem-solving, statistical reasoning, and computational thinking. Contemporary students require integrated competency development across both dimensions to succeed academically and professionally. We position these as inseparable components of workforce-ready AI orchestration.



The Evidence Base: Why Structured GenAI Integration Matters

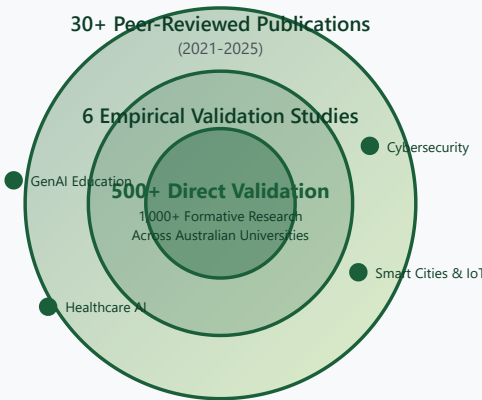
Over three years of systematic research involving 30+ peer-reviewed publications, six large-scale empirical validation studies, and direct observation of student learning across multiple Australian universities, a consistent pattern emerges: students are engaging with GenAI tools without the foundational competencies required for effective, ethical, and academically rigorous use. This evidence base, developed through formative research with over 1,000 students across 30+ publications and validated through six focused empirical studies involving 500+ participants, identifies three critical gaps that impede successful GenAI integration in higher education.

Three Evidence-Based Gaps in Student GenAI Competency

⚠ Critical Problems We Identified

- ✗ **Students don't know how to use GenAI effectively.** The majority classify themselves as beginners with minimal skills in prompt engineering, output refinement, or strategic AI use. They're using tools without understanding how to get quality results.
- ✗ **Students can't tell if AI output is accurate or biased.** They accept whatever GenAI produces without checking facts, verifying sources, or identifying potential bias. This is especially dangerous in data analysis and mathematical problem-solving where errors compound.
- ✗ **Students don't understand what responsible or ethical use means.** While they follow rules when told explicitly what to do, they don't understand why those rules exist or how to make ethical decisions in new situations. This creates academic integrity risks.

Our Research Foundation



SAGE: Evidence-Based Framework for Competency Development

The Structured AI-Guided Education (SAGE) framework was developed to systematically address these three research-validated gaps through structured pedagogical intervention. Grounded in educational psychology and validated through multi-institutional empirical studies involving 500+ students across six focused studies (with formative research foundation of 1,000+ students), SAGE provides a scaffolded approach to developing AI literacy and numeracy across all student cohorts.

Systematic Progression

Scaffolds learning from foundational understanding through advanced orchestration, empowering students through visible competency milestones and achievable progression pathways (Novice → Intermediate → Advanced → Expert).

Critical Evaluation

Motivates students to develop structured assessment and verification capabilities across qualitative and quantitative outputs through systematic practice in accuracy assessment and bias detection.

Workforce Integration

Connects academic skill development to professional contexts, empowering students to position GenAI competency as professional asset rather than academic compliance requirement.

Ethical Reasoning

Builds deep understanding of academic integrity principles and responsible AI use, allowing institutions to assure and celebrate student learning while maintaining ethical standards.

SAGE

The Evidence of Impact

46%

Students reaching advanced orchestration competency within one semester

56-87%

Performance improvements in AI-assisted task completion

78%

Enhancement in critical thinking skills applied to AI outputs

84%

Students developing capability in evaluating accuracy and bias

Direct validation: 500+ students across six studies; formative research foundation:
1,000+ students across 30+ publications

What Students Are Asking For

Analysis of open-ended student feedback across our research studies reveals consistent themes. Students seek empowerment through structured learning rather than restriction through prohibition.



More examples of what constitutes 'correct' versus 'incorrect' use in assignments.

Seeking Clarity

15% request examples of ethical use



Workshops on how to cite AI-generated content properly.

Requesting Structure

30% request workshops and tutorials



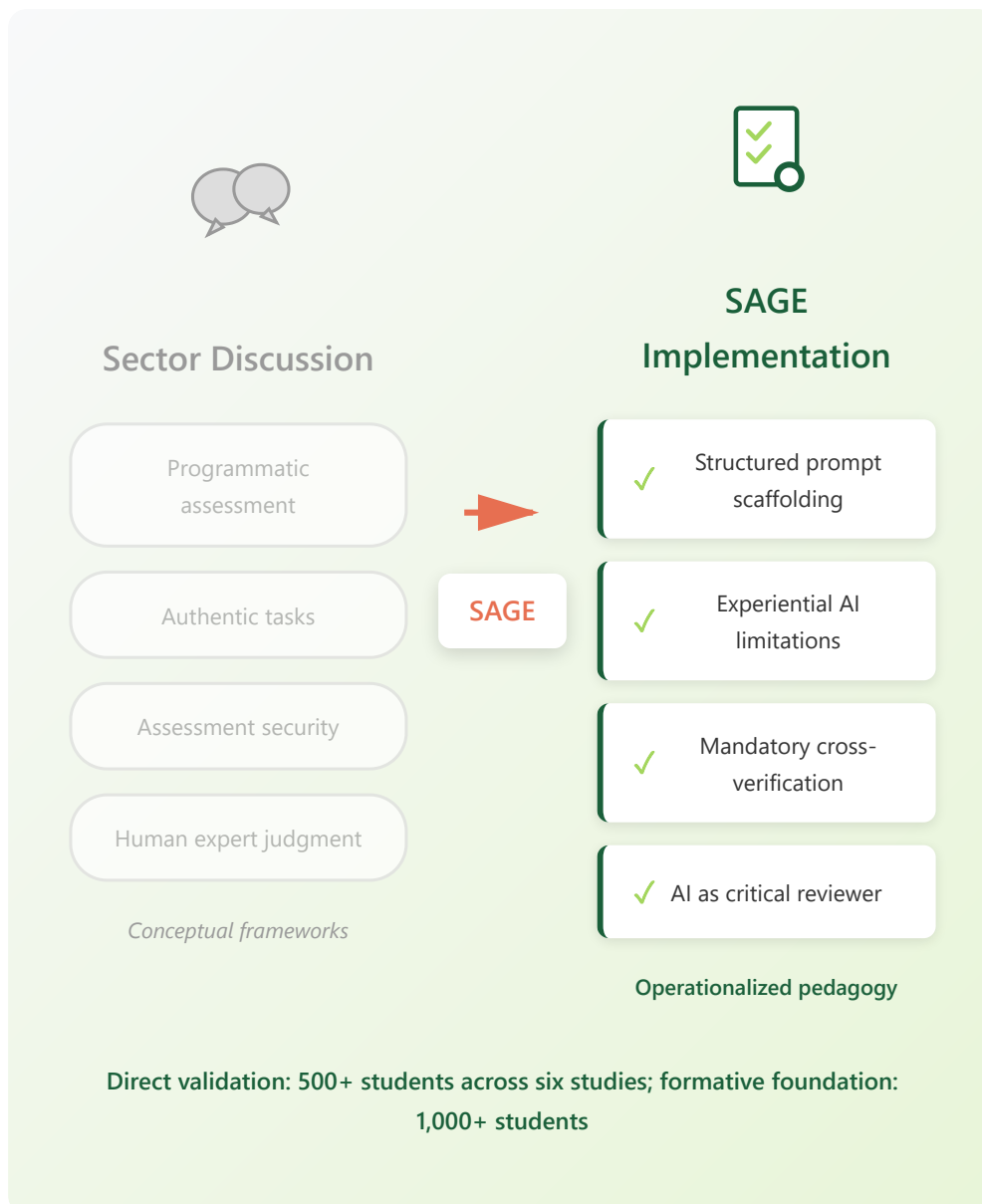
Clarify the university's stance on specific tools like ChatGPT.

Wanting Guidance

45% request clearer policies

SAGE Assessment Redesign: From Theory to Practice

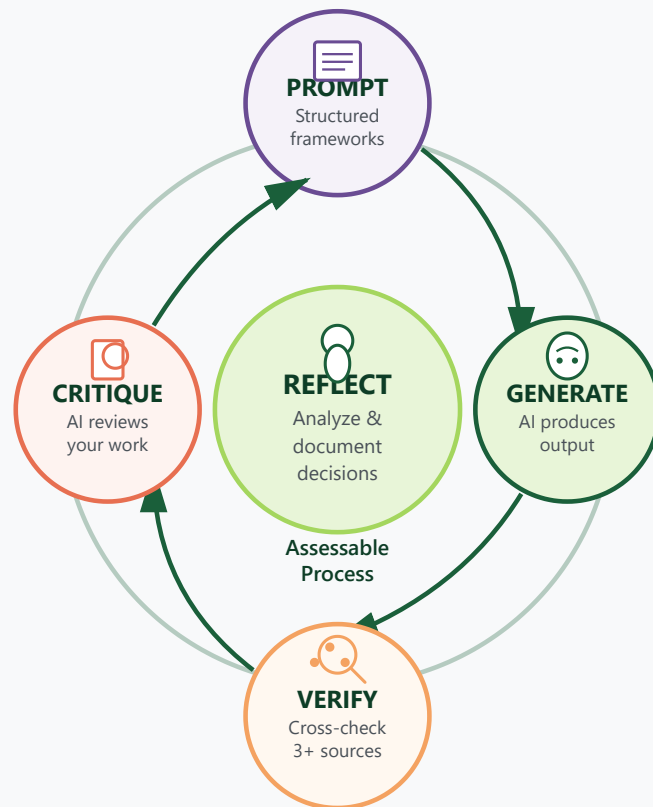
The higher education sector is actively discussing assessment redesign in response to GenAI capabilities. Leading voices advocate for programmatic assessment approaches, authentic tasks that "empty the value of cheating," and increased reliance on human expert judgment. These are valuable conceptual frameworks. SAGE moves beyond conceptual discussion to operationalized pedagogy—assessment strategies that have been implemented, refined, and validated across 500+ students in six focused empirical studies (building on formative research with 1,000+ students).



Operationalizing Assessment Redesign

Traditional assessment approaches position GenAI as either prohibited tool or unregulated resource, creating a binary that fails to develop student competency. SAGE assessment redesign fundamentally restructures the relationship between students, AI tools, and learning outcomes through four pedagogical strategies validated across multiple institutional contexts.

The SAGE Assessment Cycle



Strategy 1: Structured Prompt Scaffolding

Rather than allowing students unguided access to GenAI tools, SAGE assessments provide structured prompt frameworks that scaffold progression from novice to expert orchestration. Students receive explicit templates for initial prompts, refinement questions, and verification queries. Early assessments use highly structured prompts with clear success criteria; later assessments reduce scaffolding as competency develops. This approach makes AI orchestration itself an assessable skill rather than hidden process. By cross checking established sources and standards, students realise the shortcomings and limitations of AI and learn to verify facts before blindly trusting GenAI tools. This hands on learning approach resonate better with students than policies

Example:

In cybersecurity management assessments, students receive prompt frameworks that require them to specify context parameters, define constraints, and articulate verification requirements before generating AI outputs. This transforms prompt engineering from implicit skill to explicit learning objective.

Strategy 2: Experiential Discovery of AI Limitations

SAGE assessments create structured opportunities for students to encounter AI bias, factual errors, and reasoning limitations through guided experimentation rather than passive instruction. Assignments deliberately incorporate scenarios where GenAI tools produce flawed outputs—requiring students to identify problems, explain why errors occurred, and demonstrate correction strategies.

This pedagogical approach proves more effective than lecture-based warnings about AI limitations. Students who discover bias through structured experiential tasks demonstrate significantly higher retention of evaluation skills and more sophisticated understanding of AI capabilities and constraints. The learning emerges from doing rather than being told.

Example:

Students analyze AI-generated policy recommendations on cybersecurity frameworks, discovering that the AI conflates standards from different jurisdictions or misapplies compliance requirements. The assessment requires documentation of specific errors, explanation of why they represent problems, and corrected versions with proper source verification.

Strategy 3: Mandatory Cross-Verification Protocols

SAGE assessments embed systematic verification requirements that make cross-checking AI outputs against authoritative sources a core assessment component. Students must demonstrate three-source verification for factual claims, policy framework alignment for recommendations, and peer-reviewed research support for technical assertions. The verification process itself generates assessable artifacts—annotated AI outputs showing source checking, comparison matrices documenting discrepancies, and reflection statements explaining verification methodology.

This approach addresses the critical evaluation gap by making verification visible, assessable, and integral to task completion rather than optional quality enhancement. Students cannot achieve high grades by submitting unverified AI outputs, regardless of surface plausibility. More importantly it teaches them a job ready-important skill: verify AI output before adoption. A skill indeed becoming extremely important in the industry

Example:

When students use GenAI to analyze threat landscapes, assessments require citation of specific threat intelligence reports, alignment with frameworks like MITRE ATT&CK, and cross-reference with current vulnerability databases. The assessment rubric allocates significant weight to verification quality, not just final output quality.

Strategy 4: AI as Critical Peer Reviewer

SAGE assessments leverage GenAI tools as critical reviewers of student work, requiring students to submit their outputs to AI critique, analyze the feedback received, and document response decisions. This creates metacognitive opportunity where students must evaluate AI evaluation—determining which criticisms are valid, which reflect AI limitations, and which require human judgment to resolve.

This strategy develops sophisticated understanding of AI capabilities while building reflective practice skills. Students learn AI tools can identify surface-level errors (grammar, citation format) more reliably than deep conceptual problems (logical coherence, theoretical alignment), developing nuanced understanding of when to trust AI feedback versus seeking human expert review.

Example:

Students submit draft cybersecurity risk assessments to AI tools for critique, receive feedback on methodology and recommendations, then write reflective analysis explaining which AI suggestions they accepted (with justification), rejected (with reasoning), or flagged for instructor review (explaining uncertainty). The reflection demonstrates critical engagement with AI feedback rather than passive acceptance.



Structured Prompt Scaffolding

"Make orchestration visible"

Explicit learning objective with progressive complexity



Experiential Discovery

"Learning through doing"

Higher retention than lecture-based instruction



Cross-Verification Protocols

"Make verification assessable"

Three-source minimum with documented process



AI as Critical Reviewer

"Metacognitive development"

Evaluate the evaluator - build reflective practice

Structural Redesign: Continuous Competency Development

SAGE assessment architecture replaces single high-stakes tasks with sequenced assignments building progressive competency. Early assessments focus on basic skills (structured

prompting, simple verification) with extensive scaffolding and formative feedback. Mid-semester assessments increase complexity while reducing scaffolds. Final assessments require independent orchestration with sophisticated verification across complex scenarios.

What Gets Assessed: The Complete Picture

Traditional Assessment Focus

Final written output

20%

Surface correctness

visible

Traditional Assessment Line

SAGE Assessment Focus

Prompt engineering decisions

Source verification process

Error identification capability

AI critique evaluation

Reflective reasoning

Iteration decisions

80% of meaningful learning

SAGE assesses the 80% that traditional approaches miss

This programmatic approach allows students multiple opportunities to develop and demonstrate competency while providing continuous evidence of learning progression. Assessment becomes learning mechanism rather than summative judgment.

Why This Works: Evidence from Implementation

These assessment strategies have been validated across four empirical studies involving diverse student cohorts in metropolitan, regional, and online contexts. The approaches prove effective because they:

- Make AI orchestration skills explicit and assessable rather than hidden
- Create genuine learning opportunities where failure is informative rather than catastrophic
- Align assessment tasks with authentic professional contexts where AI verification is standard practice
- Reduce academic integrity risks by requiring demonstrated process, not just final products
- Develop transferable skills applicable beyond specific course content

Traditional assessment asks: "What did you produce?" SAGE assessment asks: "How did you produce it, how did you verify quality, and what did you learn from the process?" This shift from product to process-plus-product fundamentally changes the student-AI relationship from one of potential deception to one of sophisticated orchestration.

Distinguishing Theory from Practice

The assessment strategies described above are not aspirational proposals—they represent current practice validated through six focused empirical studies involving 500+ students (building on a formative research foundation with 1,000+ students). While the sector discusses what assessment redesign might look like, SAGE provides documented evidence of what works when theory meets classroom implementation. The framework offers practical guidance for educators seeking to move beyond conceptual discussion to actionable assessment transformation that empowers students, motivates engagement, and assures learning outcomes.

SECTION 1

Purpose and What This Resource Offers

This resource provides a field-tested, research-informed and discipline-agnostic method, **Structured AI-Guided Education (SAGE)**, for integrating generative AI responsibly into teaching, learning, and assessment whilst preserving academic standards, integrity and students' critical thinking.

What You Get

- ✓ **A clear framework** — Understand SAGE in 10 minutes
- ✓ **6-step implementation blueprint** — Start next term with confidence
- ✓ **Assessment patterns and rubrics** — Ready to adapt to your discipline
- ✓ **Templates** — Copy-paste decision logs, scaffolds, and marking criteria
- ✓ **Monitoring indicators** — Track what works and adjust quickly
- ✓ **Risk controls** — Maintain academic integrity whilst enabling AI use

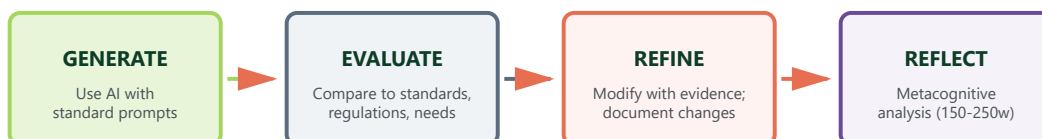


Evidence Base

Every recommendation is grounded in empirical results from **six studies across three disciplines** (cybersecurity, data analytics, systems analysis) involving **500+ students** at undergraduate and postgraduate levels across **four Australian campuses**.

The SAGE 4-Step Cycle

At the heart of SAGE is a simple, repeatable cycle that moves students from passive AI consumption to critical AI orchestration:



Key Insight

This cycle repeats at increasing complexity: first in **scaffolded tutorials** (formative), then in **authentic assessments** (summative). Students learn to accept, modify, or reject AI outputs with explicit, evidence-based reasoning.

Evidence at a Glance

500+

STUDENTS

6

STUDIES

4

CAMPUSES

3

DISCIPLINES

Key Findings That Inform SAGE

- ✓ **+56% to +87% performance improvement** with structured AI use vs. search engines
- ✓ **46% of students reached Balanced Integrator level** (thoughtful accept/modify/reject decisions) in one semester
- ✓ **AI systematically fails at boundaries, state management, and exceptions** — scaffolds address these predictable weaknesses
- ✓ **Accessibility drops from 85% → 10% → 90%** across abstraction layers without explicit prompts
- ✓ **Justification quality strongly predicts synthesis evidence** ($\rho=0.84$) — mark the process, not just the product

Who Should Use This Resource

Unit Coordinators

Redesigning assessments to incorporate GenAI whilst maintaining academic standards

Course Designers

Embedding AI literacy as a graduate attribute across programs

Learning Designers

Supporting faculty with scaffolds, rubrics, and assessment templates

Academic Integrity Leads

Balancing policy with pedagogical practice; enabling responsible use

Before You Begin

SAGE is designed for contexts where **AI use is permitted with appropriate scaffolding and disclosure**. If your institution currently prohibits all AI use, this guide offers a research-backed pathway to reconsider that policy. If prohibition remains in place, focus on assessment redesign that makes AI less useful (oral exams, in-class tasks, reflective portfolios).

What Comes Next

The remainder of this guide is structured to move you from understanding to implementation:

- **Section 2:** The evidence base — detailed findings from six studies
- **Section 3:** The SAGE framework explained — two-stage model, competency dimensions, orchestration levels
- **Section 4:** Implementation blueprint — six steps you can follow next term
- **Section 5:** Assessment patterns — three ready-to-adapt designs
- **Section 6:** Templates and rubrics — copy-paste resources
- **Section 7:** Discipline-specific examples — quick-start for your field
- **Section 8:** Academic integrity controls — pragmatic risk mitigation
- **Section 9:** FAQs and troubleshooting
- **Section 10:** Further resources and support

Estimated reading time: 25-30 minutes complete | 10 minutes for framework overview
(Sections 1-3)

SECTION 2

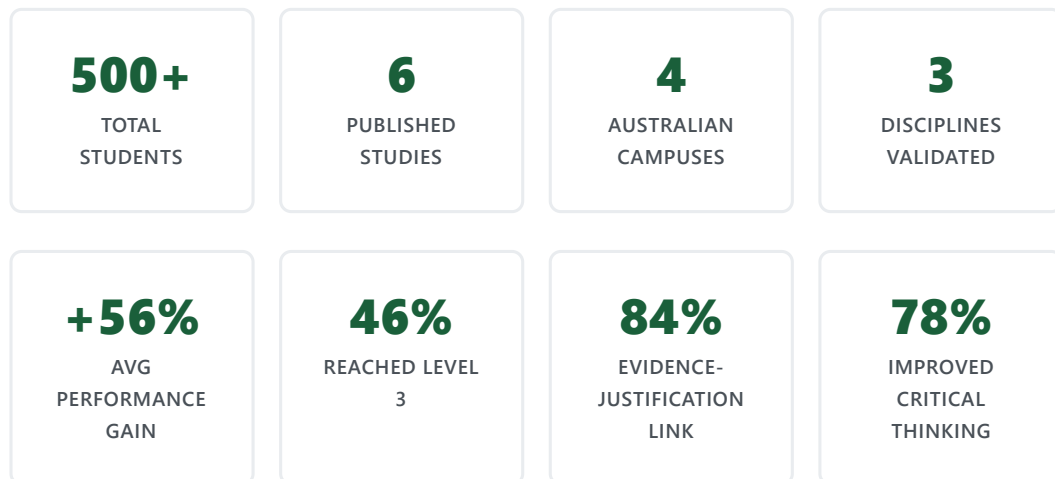
Evidence Snapshot

SAGE is grounded in empirical research across multiple contexts, student cohorts, and disciplines. This section provides key findings at a glance; comprehensive methodology and detailed results are available in Section 5.2 for institutional validation purposes.

Research Portfolio: Six Validated Studies

STUDY	SAMPLE	METHOD	KEY FINDING
Study 1 ICT Students & ChatGPT	52 students (UG + PG) 3 cohorts	Controlled comparison: ChatGPT vs search engines	56–87% performance improvement in functionality and information design with structured AI use
Study 2 Data Analytics	74 students (73% UG, 27% PG) Business + ICT	Hands-on tutorial + 27-item survey	Large effect size ($\eta^2=0.173$) for perceived benefits; satisfaction 3.88/5; concerns about complexity handling
Study 3 Cybersecurity Management	105 students (72 UG, 33 PG) 12-week semester	Two-stage SAGE implementation with 3 assessments	90%+ completion; 78% reported improved critical thinking; identified patterns in AI policy gaps
Study 4 Systems Analysis & Design	18 groups 4 campuses (Brisbane, Melbourne, Sydney, Online)	Three embedded experiments (requirements, DFD, design evaluation)	46% reached Balanced Integrator; 38% Selective Adapter; identified accessibility U-curve and systematic AI failures
Study 5 AI as Critic	~60-70 students Systems Analysis & Design	AI as critical evaluator (reversing AI-as- generator pattern)	Strengthened student ownership; validated human authority principle; disciplined accept/reject decisions
Study 6 From Anxiety to Agency	~160+ students ICT Programs	Mixed-methods examining student ethical frameworks	Significant reduction in AI anxiety; increased sense of agency; student- driven ethical integration frameworks

Key Outcomes at a Glance



What the Research Reveals

✓ Evidence of Effectiveness

- Performance improvements of 56-87% with structured AI use
- 84% of students reached Level 2 or 3 (Selective Adapter or Balanced Integrator) within one semester
- Strong correlation ($p=0.84$) between justification quality and evidence use
- Students confidently reject context-inappropriate AI advice (90% in design tasks)



Challenges Requiring Scaffolding

- 15% remained Passive Acceptors without intervention
- 55% struggled with boundary reasoning; 45% with state management
- Accessibility dropped from 85% → 10% → 90% without explicit prompts
- AI systematically failed at context, compliance, and exceptions

Research-to-Practice Translation

Every SAGE recommendation addresses a specific pattern observed across these six studies. The assessment examples in Section 4 incorporate scaffolds that mitigate the challenges identified (boundary prompts, accessibility checkpoints, mandatory rejection quotas). The framework is not theoretical but a distillation of what worked across 1000+ students in authentic educational settings. **Detailed methodology and findings are available in Section 5.2** for educators seeking institutional approval or conducting their own research validation.

SECTION 3

The SAGE Framework in Detail

Core Proposition

*Treat GenAI as an **assistive tool whose outputs must be accepted, modified, or rejected with explicit justification**—not as an authority to mimic nor as a tool to ban.*

What is SAGE? A Simple Explanation

SAGE teaches students to "orchestrate" AI outputs rather than blindly accept them.

Think of a conductor leading an orchestra: they don't play every instrument themselves, but they decide which instruments to feature, when to adjust the tempo, and when something sounds wrong. Similarly, students learn to use AI as one tool among many accepting its strengths (speed, initial drafts), correcting its weaknesses (missing context, regulatory gaps, boundary errors), and knowing when to override it entirely based on professional judgement.

In practice, this means students don't just ask AI "write me a policy" and submit what it produces. Instead, they generate an AI draft, compare it to industry standards and regulations, identify what's missing or wrong, refine it with evidence from research and stakeholder input, and justify every decision they make. The AI does the initial heavy lifting; the student does the critical thinking.

What "Orchestration" Means in Practice

Every time a student uses AI in SAGE, they must make explicit **Accept / Modify / Reject** decisions:

✓ Accept

"This AI recommendation is appropriate because it aligns with [standard/regulation]. I verified it against [source]."

🔄 Modify

"AI suggested X, but context Y requires Z instead. I changed it because [evidence-based reason]."

✗ Reject

"AI recommended X, but this is inappropriate for our setting because [contextual reason]. I'm removing it and replacing with [alternative]."

This forces students to think critically about *why* each AI suggestion does or doesn't work, rather than passively copying.

How SAGE Works: Two Quick Examples

Here's what SAGE looks like in practice across two different disciplines:

Example 1: Cybersecurity Policy Development

Context: Third-year students developing a BYOD (Bring Your Own Device) policy for a healthcare provider

Student's Prompt to ChatGPT:

"Create a BYOD policy for a 200-person healthcare clinic that handles patient records. Include device security, data protection, and staff responsibilities."

What AI Produces:

- Generic "use strong passwords" guidance
- Basic encryption recommendation
- Employee training mentioned but not detailed

What AI Misses:

- **Privacy Act 1988 (APP 11.1):** Specific requirements for health data protection
- **Enforcement mechanisms:** What happens if staff violate policy?
- **Third-party vendor risks:** Personal devices accessing cloud services
- **Incident response:** What to do if device is lost/stolen?

Student's SAGE Response:

MODIFY: Changed "use strong passwords" to "mandatory 12+ character passwords with MFA enforced via MDM (Mobile Device Management), audited quarterly"

Justification: Privacy Act APP 11.1 requires "reasonable steps" for health data; NIST SP 800-63B specifies MFA for sensitive data

REJECT: Removed AI's suggestion for "optional encryption"

Justification: Encryption must be mandatory, not optional, for health records under Australian Privacy Principles

ADD (AI didn't include): Remote wipe capability must be enabled on all devices; 24-hour reporting requirement for lost/stolen devices; progressive discipline policy (warning → suspension → termination)

Justification: Based on interview with clinic's IT manager and ISO 27001 control A.6.2.1 (mobile device policy)

Example 2: Nursing Care Plan for Elderly Patient

Context: Second-year nursing students developing discharge plan for 78-year-old with limited mobility

Student's Prompt to ChatGPT:

"Create a discharge care plan for a 78-year-old patient with hip fracture, limited mobility, living alone. Include medication management, fall prevention, and follow-up care."

What AI Produces:

- Standard fall prevention checklist (remove rugs, install grab bars)
- Medication schedule template
- Generic follow-up appointment recommendation

What AI Misses:

- **Home environment specifics:** Does patient have stairs? Can they access bathroom safely?
- **Social support:** Who will help with shopping, cooking, transport?
- **Cultural considerations:** Indigenous health protocols, language barriers, family decision-making preferences
- **Cognitive status:** Can patient manage medication independently?

Student's SAGE Response:

ACCEPT: Generic fall prevention items (grab bars, non-slip mats, adequate lighting)

Justification: Aligned with NSW Health Fall Prevention Guidelines; verified during home safety assessment

MODIFY: Changed "patient will self-administer medications" to "Webster pack arranged via pharmacy; daughter to visit daily for first 2 weeks to supervise"

Justification: Patient scored 22/30 on MMSE (mild cognitive impairment); family assessment identified daughter as primary support; aligns with Aged Care Quality Standards (3.2 - personal care and clinical care)

REJECT: Removed AI's suggestion for "online telehealth follow-ups"

Justification: Patient has no smartphone/computer and limited digital literacy; in-person home nursing visits more appropriate

ADD (AI didn't include): Referral to Aboriginal Liaison Officer (patient identified as Indigenous); arrange interpreter for follow-up (patient's first language is Warlpiri); coordinate with Community Aged Care Package for meal delivery and transport to appointments

Justification: Cultural Safety Framework requires Indigenous-specific support; NDIS Practice Standards mandate culturally appropriate care

Key Insight from Examples

Notice the pattern: AI provides a **useful starting point** (generic best practices), but students must **add context-specific details** (regulations, stakeholder needs, cultural considerations, environmental constraints). The AI saves time on initial drafting; the student adds critical thinking, domain expertise, and professional judgement. **This is orchestration- an industry required skill now taught to students**

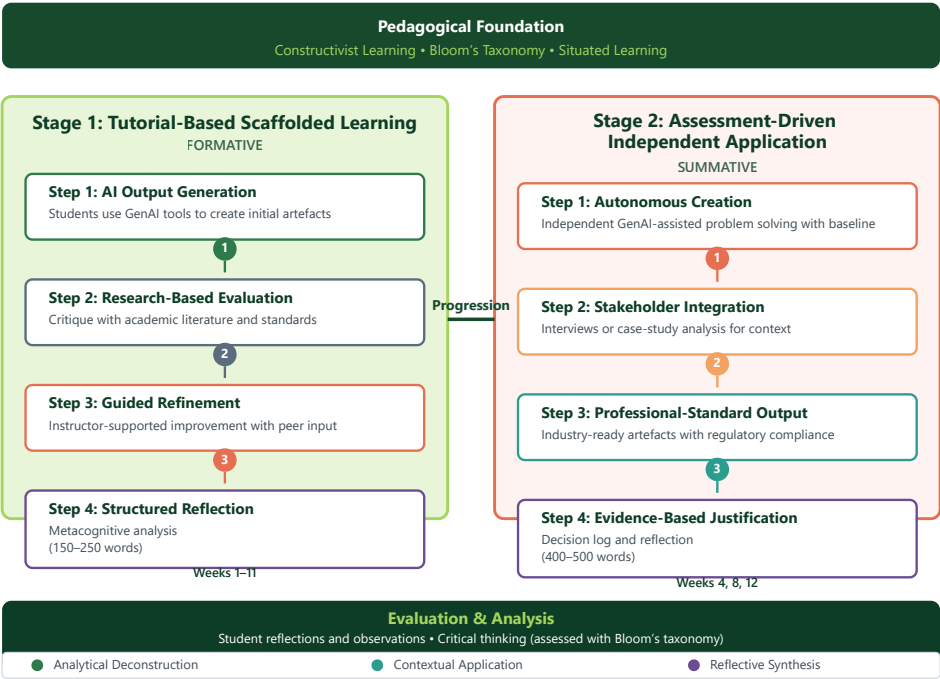
The SAGE Framework Model

SAGE operates through a two-stage progression, moving from **guided practice** (formative) to **independent application** (summative).

Why SAGE Uses Experiential Learning

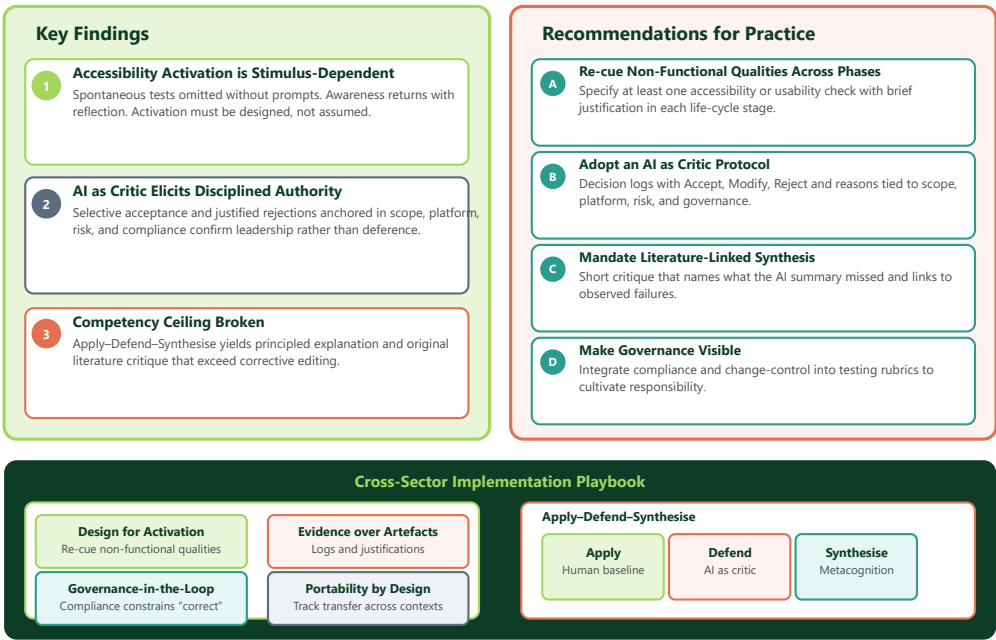
SAGE assumes that students require scaffolded guidance when learning to work with AI, not abstract warnings about its limitations. Rather than instructing students that "AI may generate culturally insensitive content" or "AI outputs can be generic and lack context," SAGE provides structured prompts that lead students to **discover these limitations through experience**. When a nursing student realises their AI-generated discharge plan failed to consider Indigenous health protocols, or when a cybersecurity student discovers AI omitted jurisdiction-specific regulations, the lesson is internalised far more effectively than any lecture could achieve. This experiential approach—allowing students to encounter AI's strengths and weaknesses firsthand through carefully designed tasks—ensures the learning is both memorable and transferable to professional practice.

SAGE Framework: Two-Stage Progression



SAGE Findings and Recommendations for Practice

Orchestration Competencies • Accessibility Activation • AI as Critic • Metacognitive Synthesis



What Students Actually Do in Each Step

Here's what the framework looks like in practice:

Stage 1: Tutorial Practice (Weeks 1-4)

Step 1 — Generate:

You give students a standardised prompt (e.g., "Ask ChatGPT to create an email security policy for a small business"). They paste it into AI, capture the output as a screenshot, and submit it. This ensures everyone starts with comparable AI responses.

Step 2 — Evaluate:

Students compare the AI output to materials you provide—a checklist of NIST controls, a sample policy from industry, or relevant legislation. They mark what's present, what's missing, and what's wrong. You might provide a template: "Does it address multi-factor authentication? Yes/No. Does it specify password complexity? Yes/No."

Step 3 — Refine:

Students revise the AI output based on their evaluation. They add missing requirements, correct errors, and align it with the standards. They use track changes or side-by-side comparison so you can see what they changed. Peer review or tutor feedback helps at this stage.

Step 4 — Reflect:

Students write 150-250 words answering: "What did AI do well? What did it miss or get wrong? When should you trust AI vs verify it yourself?" This metacognitive step ensures they're not just correcting mechanically—they're thinking about AI's patterns of strength and weakness.

Stage 2: Independent Assessment (Weeks 5-12)

Step 1 — Autonomous Creation:

Students design their own prompts and use AI without your scaffolding. They must first create a human baseline (bullet points, hand-drawn diagram, interview notes) before using AI—this proves they understand the problem independently. They submit both the baseline and AI output.

Step 2 — Stakeholder Integration:

Students engage with real-world complexity: interview a business client, analyse a detailed case study with constraints and trade-offs, or review actual organisational policies. They identify gaps between what AI produced and what stakeholders actually need. This is where context matters most.

Step 3 — Professional Output:

Students produce an industry-ready artefact—a policy that could be implemented, a design that meets accessibility standards, a data model that satisfies regulatory requirements. The quality standard is "ready for client presentation," not "good enough for a student assignment." AI can help polish, but students must ensure correctness.

Step 4 — Evidence-Based Justification:

Students submit a comprehensive decision log (400-500 words) documenting every Accept/Modify/Reject choice. For each decision, they cite evidence: "I modified AI's recommendation because Privacy Act 1988 requires X" or "I rejected AI's suggestion because stakeholder Y indicated Z." This is where you assess their orchestration competence.

How to Read This Framework

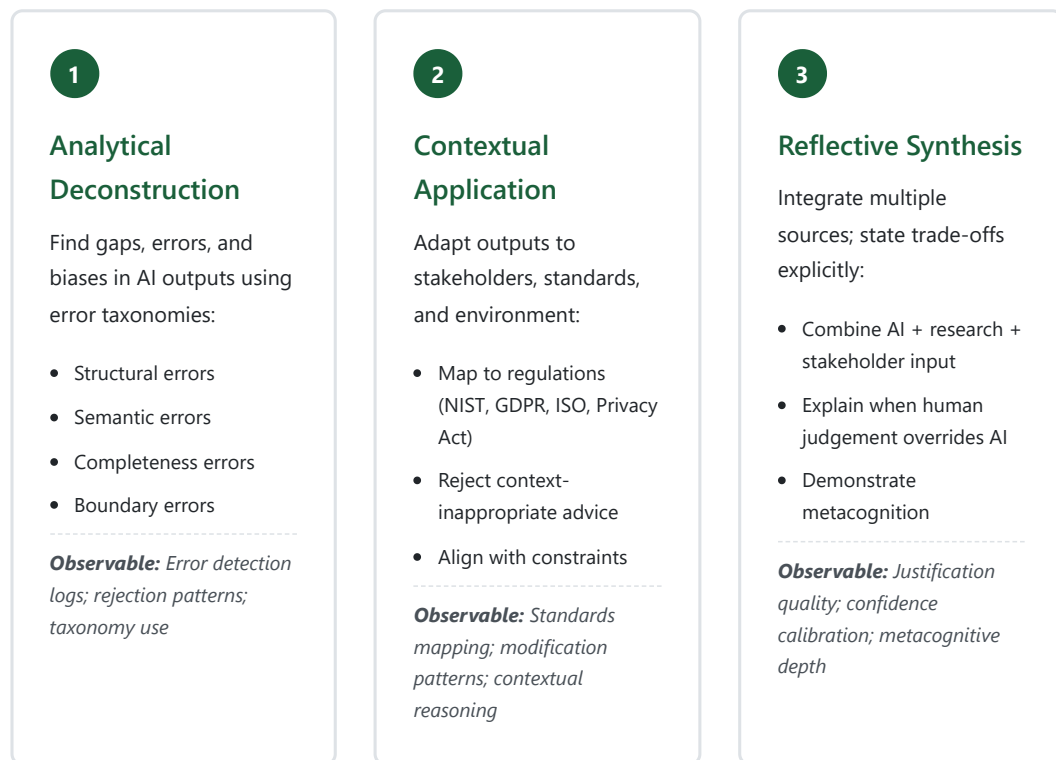
Left side (green): Weeks 1-4, students practice AI evaluation in tutorials with heavy scaffolding—you provide templates, worked examples, and structured feedback.

Right side (coral): Weeks 5-12, students apply AI independently in assessments with reducing scaffolding—they must produce side-by-side artefacts (human baseline + AI output + refined version + justification).

Bottom (dark green): Throughout, you evaluate their orchestration process using Bloom's taxonomy and student reflections, not just the final output quality.

Three Observable Competency Dimensions

SAGE operationalises AI orchestration through three Bloom L3-L6 competency dimensions that can be observed and measured:



Student Orchestration Levels

SAGE moves students through four measurable levels of AI orchestration competence. These levels were formalised from empirical analysis in our cybersecurity management study (Study 3), where patterns emerged from observing how students interacted with AI-generated policy content across multiple assessments. Our study (6) also showed how students interact with AI. The taxonomy distinguishes students based on three observable criteria:

Criteria for Classification

1. **Decision distribution:** The ratio of Accept/Modify/Reject decisions when evaluating AI outputs (e.g., accepting 50% vs 90% indicates different critical engagement levels)
2. **Justification depth:** Whether students provide evidence-based reasoning with specific citations to standards, regulations, or research—or rely on generic statements like "AI was incomplete"
3. **Contextual reasoning quality:** The ability to override AI based on organisational constraints, stakeholder needs, boundary considerations, or domain-specific requirements that AI cannot know

These criteria were operationalised through rubric-based assessment of student artefacts (decision logs, reflection essays, and side-by-side comparisons) and validated across multiple cohorts and campuses in subsequent studies. Our research shows most students reach Level 3 within one semester:

Level 4	Critical Synthesiser — Proactively finds gaps AI cannot address; generates novel solutions beyond AI; high metacognitive sophistication	0%*
Level 3	Balanced Integrator — Accepts ~50%; modifies ~40%; rejects ~10%; evidence-based justifications; demonstrates boundary reasoning	46%
Level 2	Selective Adapter — Accepts ~70%; modifies ~25%; rejects ~5%; surface reasoning; struggles with boundaries and exceptions	38%
Level 1	Passive Acceptor — Accepts >90%; minimal critique or modification; justifications generic or absent	15%

* Level 4 requires advanced scaffolding (adversarial/gap-finding assessments) not included in initial studies



Implementation Goal

Target for one-term implementation: Move 70%+ of students to Level 3 (Balanced Integrator) by end of semester. Reaching Level 4 (Critical Synthesiser) requires deliberate scaffolding with open-ended, gap-finding assessments.

The "Critical Consideration Drop" Problem

Across multiple assessments in our systems analysis study (Study 4), we identified a troubling pattern: **important design considerations that students articulated clearly in early stages disappeared during mid-level work, then re-emerged at final stages.** This creates a U-shaped curve of awareness—high at the beginning, critically low in the middle, recovering at the end.

Why This Matters Across Disciplines

Every discipline has critical considerations that must be carried through the entire workflow; yet often get "lost" during technical, analytical, or complex middle phases. In software design, it's **accessibility and inclusive design**. In nursing, it's **cultural safety and patient dignity**. In history, it's **Indigenous perspectives and historiographical bias**. In business, it's **ethical implications and stakeholder equity**. When students work on complex technical tasks, these essential considerations fade from attention unless explicitly scaffolded at every stage.

The Accessibility U-Curve: A Representative Example

In our systems analysis study, we tracked how often students explicitly addressed accessibility (designing for users with vision, hearing, motor, or cognitive differences) across three task types:

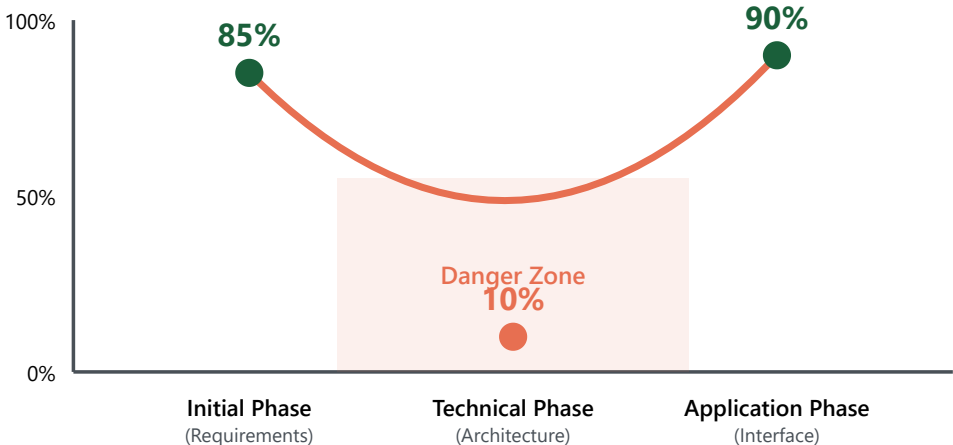
Task Progression

- **Requirements (prose):** Students interview stakeholders, write user stories
- **Architecture (diagrams):** Students create data flow diagrams, entity-relationship models
- **Interface (design):** Students design screens, workflows, interactions

Accessibility Articulation

- **85% of students** mentioned accessibility needs, assistive technology, or inclusive design
- **10% of students** included accessibility in diagrams or architectural decisions
- **90% of students** re-addressed accessibility in interface design (buttons, labels, navigation)

Critical Consideration Articulation Across Task Complexity



⚠️ Why the U-Curve Happens

Three mechanisms drive this pattern across disciplines:

1. **Cognitive salience:** Critical considerations feel relevant during initial planning (talk to stakeholders, identify needs) and final application (real users will interact with this), but seem "distant" during technical middle work (diagrams, models, analysis)
2. **Representational affordance:** Some work formats lack obvious places to encode critical considerations e.g., data flow diagrams don't have "accessibility boxes," literary analysis templates don't prompt for Indigenous perspectives, financial models don't include ethical impact cells
3. **Cognitive load:** Complex technical work (diagramming, calculating, modelling, analysing) crowds out secondary concerns unless explicitly prompted

Cross-Disciplinary Examples of U-Curve Risk

Nursing: Cultural Safety

High (85%): Initial assessment—students identify patient's cultural background, language, family structure

Low (10%): Care planning—students focus on clinical protocols, medication schedules, wound care; cultural needs disappear

High (90%): Discharge planning—students remember to involve family, arrange interpreter, coordinate with community services

History: Indigenous Perspectives

High (85%): Topic selection—students acknowledge need to include Indigenous voices, oral histories

Low (10%): Source analysis—students focus on colonial archives, government documents; Indigenous sources overlooked

High (90%): Final essay—students remember to critique colonial bias, include Indigenous historians' work

Business: Ethical Implications

High (85%): Problem definition—students identify stakeholders, ethical concerns, community impact

Low (10%): Financial modelling—students focus on ROI, cost-benefit, risk analysis; ethics fade

High (90%): Recommendations—students remember to address environmental impact, labour practices, equity

Engineering: Safety & Sustainability

High (85%): Design brief—students identify safety standards, environmental impact, lifecycle considerations

Low (10%): Technical calculations—students focus on load analysis, material properties, efficiency; sustainability drops

High (90%): Final report—students remember to address carbon footprint, disposal, maintenance

✓ SAGE Solution: Flatten Your Discipline's U-Curve

Step 1: Identify your critical consideration

What essential element must persist through all stages of student work but risks disappearing during technical/complex phases? (Accessibility, cultural safety, ethics, sustainability, Indigenous perspectives, patient dignity, etc.)

Step 2: Insert explicit prompts at every task phase

Add two mandatory questions to every assessment task, including the technical middle phases:

- "How does [critical consideration] apply to this component/phase/analysis?"
- "What evidence demonstrates you've addressed [critical consideration] at this stage?"

Step 3: Adapt tools and templates

Add explicit representation in your discipline's technical formats:

- **Software:** Add "AccessibilityProfile" data store; "AssistanceOrchestrator" process
- **Nursing:** Add "Cultural Safety Checklist" to care plan templates
- **History:** Add "Indigenous Sources" column to evidence tables
- **Business:** Add "Ethical Impact" row to decision matrices
- **Engineering:** Add "Sustainability Metrics" to calculation sheets

Step 4: Award marks at every phase

Include your critical consideration in rubrics for initial, middle, AND final work—not just at the beginning and end. If it's worth 10% at initial phase and 10% at final phase, allocate 10% in the middle phase too. Students prioritise what's marked.



Evidence from Implementation

When we added micro-notation for accessibility and two mandatory prompts to architectural tasks in Study 4's second iteration, accessibility articulation in the middle phase increased from 10% to 76%. **The pattern is generalisable:** Whatever critical consideration matters in your discipline, make it visible and assessable at every stage, especially during complex technical work where cognitive load is highest.

Connecting Framework to Your Discipline

The pattern is the same across all disciplines: Students generate AI outputs → compare to domain standards (regulations, clinical guidelines, engineering codes, business frameworks) → identify gaps → refine with evidence → justify decisions. The *specifics* change (Privacy Act vs NDIS Standards vs AS 1170), but the *process* remains consistent. This makes SAGE adaptable to cybersecurity, nursing, engineering, business, education, and beyond.

SECTION 4

Implementation Examples: SAGE in Practice

This section presents actual assessment materials from cybersecurity and systems analysis courses, showing exactly how SAGE operates in practice. Each example includes the task description, standardised prompts, templates, and mapping to SAGE framework elements.



How to Use These Examples

These are **copy-paste ready materials**. Review the complete example in your discipline area (or closest match), then use the adaptation guide (Section 4.3) to modify prompts, standards, and context for your field. The underlying SAGE structure remains constant—only the domain specifics change.

4.1 Formative Example: Email Policy Development Tutorial

Discipline: Cybersecurity Management | **Level:** Undergraduate/Postgraduate | **Duration:** 2-hour tutorial | **SAGE Stage:** Stage 1 (Formative, High Scaffolding)

Context and Learning Objectives

Students develop an email security policy for a fictional bank (XYZ Bank) by generating an initial AI draft, refining it using academic research and industry frameworks, then reflecting on the process. This tutorial operationalises the complete SAGE 4-step cycle in a single session.

SAGE Framework Mapping

SAGE Elements Present

- ✓ **Generate:** Standardised prompt ensures comparable AI outputs
- ✓ **Evaluate:** Students compare AI policy to research paper + SANS templates
- ✓ **Refine:** Guided improvement using specific research findings (7 scaffolded areas)
- ✓ **Reflect:** 300-word metacognitive analysis comparing AI vs human-developed policy

Competency Dimensions Developed

- ① **Analytical:** Identify what AI missed (enforcement, vendor risk, training)
- ② **Contextual:** Map to Privacy Act 1988, ISO 27001, NIST frameworks
- ③ **Reflective:** Articulate AI's role in policy development

The Complete Tutorial Task

Part 1: Use GenAI to Create an Initial Policy

Instructions to students:

- Use a Generative AI tool (ChatGPT, Bard, or Claude) to generate an Email policy for XYZ Bank
- Copy and paste the AI-generated policy into your submission
- **Do not edit the AI-generated response at this stage**

No standardised prompt provided here—students experiment freely. This reveals AI's "natural" output quality and sets baseline for comparison. In Stage 2 assessments, standardised prompts ensure fairness.

Part 2: Read Research Paper and Develop Your Own Policy

Required reading: "Analysis of the Human Factor in Cybersecurity: Identifying and Preventing Social Engineering Attacks in Financial Institutions" by Ibrahim Momoh et al.

Scaffolded guidance provided (7 areas):

1. Overview and Purpose

Students must outline risks from email misuse; highlight phishing awareness training (as recommended in the paper)

2. Acceptable Use and Behaviour

Establish rules prohibiting clicking links/attachments from unknown senders; clarify business vs personal use

3. Security Controls

Prohibit auto-forwarding to external addresses; **students must add two additional controls** based on paper

4. Training and Awareness

Design continuous awareness program; include phishing simulations (per the paper recommendations)

5. Incident Response Plan

Outline automated reporting tools; specify response team roles

6. Monitoring and Privacy

Inform employees of monitoring; justify using paper's findings on transparency

7. Continuous Evaluation

Include regular updates section; reference paper's recommendation on adapting to emerging threats

SAGE Design Note:

Heavy scaffolding ensures students know WHERE to look for gaps. This is Stage 1: we are teaching the evaluation skill itself. By Stage 2, students identify gaps independently without such explicit prompts. Once this skill is acquired by the students, then the scaffolding can be minimised to focus more on the task.

Additional requirement:

Use SANS security policy templates (General category) and acknowledge source

Use Harvard referencing style or any acceptable referencing style

Part 3: Reflection (300 words)

Required reflection questions:

- 1. How did the AI-generated policy differ from your own policy? In what ways did the AI policy fail to address the specific risks and operational environment at XYZ Bank, and how did your policy improve on this?
- 2. How do you view the role of AI in drafting security policies after this exercise?

SAGE Design Note:

Metacognitive reflection is mandatory in all SAGE tasks. This forces students to articulate AI's limitations explicitly rather than subconsciously noting them. Builds confidence to reject AI in future tasks.

Why This Tutorial Works

Strengths

- **Immediate discovery:** Students see AI's gaps firsthand rather than being told
- **Research integration:** Paper provides evidence for modifications
- **Template access:** SANS frameworks show professional standards
- **Low-stakes practice:** Formative tutorial allows experimentation

Typical Student Discoveries

- AI provides generic advice ("use strong passwords")
- Missing: Enforcement mechanisms, vendor risk, training detail
- No context-specific customization for banking sector
- Regulatory requirements (Privacy Act) absent

Cross-Discipline Adaptation

IF TEACHING...	SWAP XYZ BANK POLICY TO...	SWAP RESEARCH PAPER TO...	SWAP SANS TEMPLATES TO...
Nursing	Discharge care plan for elderly patient with chronic conditions	Research on cultural safety in aged care or Indigenous health protocols	NSW Health guidelines; NDIS Practice Standards; clinical pathway templates
Business	Stakeholder communication plan for organizational restructure	Research on change management or organizational behaviour during transitions	Project Management Institute templates; change management frameworks (Kotter, ADKAR)
Engineering	Safety management plan for construction site	Research on construction safety or risk management in civil engineering	Australian Standards (AS 1170); Safe Work Australia guidelines; engineering codes
Education	Inclusive curriculum plan for diverse learners	Research on Universal Design for Learning or differentiated instruction	ACARA curriculum documents; UDL guidelines; state education frameworks

Implementation Tip

The 7 scaffolded areas (Overview, Acceptable Use, Security Controls, etc.) are generic policy sections. Replace with discipline-appropriate sections:

Nursing: Assessment → Diagnosis → Planning → Implementation → Evaluation

Business: Situation → Complication → Resolution → Action

The structure stays the same; only the content changes.

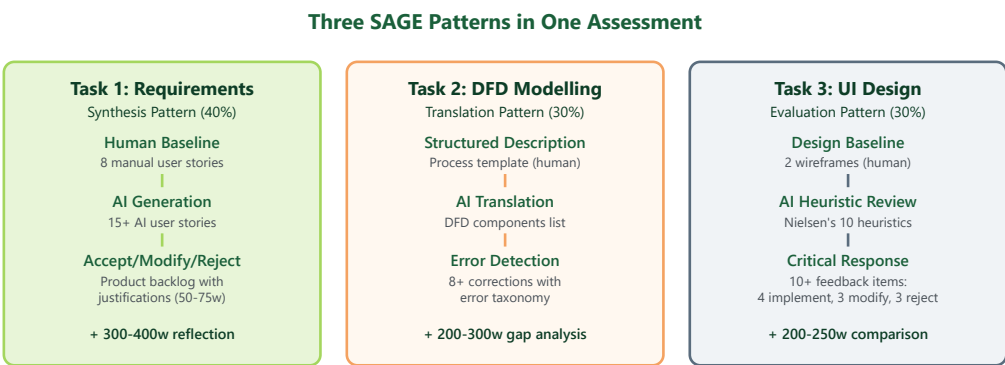
4.2 Summative Example: Agile Design and Modelling Assessment

Discipline: Systems Analysis & Design | **Level:** Undergraduate | **Weighting:** 30% | **Group Size:** 4 students
| **SAGE Stage:** Stage 2 (Summative, Reducing Scaffolding)

Assessment Overview

Students design components of a smart kiosk system for a supermarket chain (GreenHarvest) through three tasks that operationalise all three SAGE assessment patterns. The assessment explicitly frames itself as research on AI orchestration in professional practice.

SAGE Framework Mapping Across Three Tasks



Task 1: Requirements Synthesis (The Synthesiser Experiment)

SAGE Pattern: Requirements synthesis | **Orchestration focus:** Accept/Modify/Reject decisions with evidence-based justification | **Target competency:** Balanced Integrator (L3)

Part A: Manual User Story Creation

Requirement: Each team member writes 2 user stories independently (8 total per group)

Template provided to students:

As a [type of user]
I want [to perform some task]
So that [I can achieve some goal]

Acceptance Criteria:

1. [Specific measurable outcome]

- 2. [Specific measurable outcome]
- 3. [Specific measurable outcome]

Story Points: 1, 2, 3, 5, or 8

SAGE Note: Human baseline required FIRST. Prevents complete AI delegation. Creates comparison anchor.

Part B: AI-Generated Requirements

Standardised Prompt (must be used exactly):

"Based on the GreenHarvest Smart Kiosk system described below, generate comprehensive user stories for all system stakeholders.

The system includes:

- Customer product search with real-time location and availability
- Loyalty member express ordering and reorder of favourites
- Promotional offers based on customer profiles and regional trends
- Manager stock monitoring without manual inspection
- Integration readiness for future AI-driven analytics

Consider edge cases, non-functional requirements, accessibility needs for elderly users and Indigenous customers, and technical constraints. Generate at least 15 user stories in the standard format with acceptance criteria"

Required Documentation:

- Exact prompt used (if modified, explain why)
- Complete unedited AI response
- Tool name and version
- Timestamp of interaction
- Number of prompt iterations needed

SAGE Note: Standardised prompt ensures fairness and comparability. Accessibility prompts (elderly, Indigenous) surface whether AI and students both remember critical considerations from requirements → architecture → interface.

Part C: Synthesis and Critical Evaluation (Core SAGE Element)

Students create a product backlog reconciling human and AI stories using the following template:

ID	USER STORY TITLE	SOURCE	CATEGORY	PRIORITY	POINTS	CONFIDENCE (1-5)	DECISION	JUSTIFICATION
1	Quick reorder for loyalty members	Human	Core	High	5	4	Accept	Core requirement validated by stakeholder interviews. Addresses time-poor customers. Story points realistic for 2-week sprint. (52 words)

ID	USER STORY TITLE	SOURCE	CATEGORY	PRIORITY	POINTS	CONFIDENCE (1-5)	DECISION	JUSTIFICATION
2	Voice-activated search	AI	Nice-to-have	Low	8	2	Modify	AI suggested voice input but GreenHarvest operates in noisy environment. Changed to large text input with autocomplete for elderly users instead. (67 words)
3	AI-powered recipe suggestions	AI	Out-of-scope	Low	13	5	Reject	Exceeds MVP scope and timeline. Requires nutrition database integration not in current architecture. May revisit in Phase 2 post-launch. (71 words)



Required Distribution

- Include ALL 8 human-generated stories
- Include at least 5 AI-generated stories (3 accepted/modified, 2 rejected)
- Identify at least 2 stories that merge human and AI ideas
- Minimum 15 total entries

SAGE Note: Mandatory rejection quota prevents passive acceptance (Level 1). Forces discrimination. Confidence ratings surface metacognition. 50-75 word justifications ensure evidence-based reasoning, not generic statements.

Part D: Orchestration Analysis (300-400 words)

Required reflection questions:

1. What patterns did you observe in the AI's requirements generation?
2. Which types of requirements did AI consistently generate well?
3. What critical domain knowledge was essential for evaluating AI suggestions?
4. How did your team resolve disagreements about AI-generated stories?

SAGE Note: Questions target all three competency dimensions, analytical (patterns), contextual (domain knowledge), reflective (team process). This is the Stage 2 version of the formative tutorial reflection, less scaffolded, more sophisticated analysis expected.

Task 2: DFD Translation and Correction (The Translator/Corrector Experiment)

SAGE Pattern: AI-to-model translation & correction | **Orchestration focus:** Error detection using taxonomy | **Addresses research finding:** 55% struggle with boundaries; 45% with exceptions

Part B: Structured Process Description (Human Baseline)

Students complete this template BEFORE using AI:

Process Name	Loyalty Member Express Order Processing
Trigger Event	[Student describes initiating event - 25-50 words]
Preconditions	[List 3-4 conditions - 50-75 words]
Process Steps	[Numbered list of 8-10 steps - 200-250 words]
Data Inputs	[List with sources]
Data Outputs	[List with destinations]
Data Stores Accessed	[List with operation type]
Exception Handling	[2-3 error scenarios - 100-150 words]

SAGE Note: This pre-translation scaffold is the mitigation for the research finding that students and AI both fail at boundaries/CRUD/exceptions. Explicit template forces attention to these elements BEFORE AI translation.

Part C: AI Translation and Correction

Standardised Prompt:

"Convert the following structured process description into a Level-0 Data Flow Diagram. List all components in this format:

Processes:

- Process 1.0: [Name] - [Description]
- Process 2.0: [Name] - [Description]

External Entities:

- [Entity Name]: [Role/Description]

Data Stores:

- D1: [Store Name] - [Contents]

Data Flows:

- From [Source] to [Destination]: [Data Flow Name] - [Contents]

Ensure the DFD is balanced with a context diagram that has [list your external entities] as external entities."

Error Detection and Correction Log

Students complete this table for at least 8 corrections:

ERROR ID	COMPONENT TYPE	AI OUTPUT	CORRECT VERSION	ERROR CATEGORY	COMPLEXITY	EXPLANATION (30-50 WORDS)
E001	External Entity	"Payment Gateway" as Process 2.0	Payment Gateway as External Entity	Boundary	Moderate	AI incorrectly placed third-party payment processing inside system scope. Payment Gateway is

ERROR ID	COMPONENT TYPE	AI OUTPUT	CORRECT VERSION	ERROR CATEGORY	COMPLEXITY	EXPLANATION (30-50 WORDS)
						external—system sends payment request and receives confirmation, but doesn't control processing.
E002	Data Flow	Missing "Update Customer Profile" flow	Add flow from Process 1.0 to D1 (Customer)	Completeness (CRUD)	Simple	AI generated Create and Read operations for Customer store but omitted Update. Customers must modify contact details per functional requirement FR-12.
E003	Process	No exception handling for "out of stock" scenario	Add Process 3.0: Handle Stock Exception	Completeness (Exception)	Complex	AI didn't model error pathways. When item unavailable, system must offer substitutes or remove from order —requires separate process with decision logic and user notification.

Error Categories (Empirically Derived from Study 4)

- **Structural (30%):** Incorrect flow directions, processes feeding back to themselves
- **Completeness (27%):** Missing CRUD operations, omitted data stores
- **Semantic (26%):** Misunderstood business logic, conflated entities
- **Notation (15%):** Incorrect symbols, violated DFD conventions
- **Boundary (55% of students struggle):** Scope definition errors

SAGE Note: Providing the empirically-derived taxonomy teaches students what to look for. This is Stage 2, but we're still scaffolding the error detection skill—just less than Stage 1.

Task 3: UI Design with AI Critique (The AI Critique Experiment)

SAGE Pattern: Design evaluation with heuristics | **Orchestration focus:** Contextual judgement—when to override AI based on user/environment constraints | **AI role:** Simulated usability consultant

Part A: Wireframe Design (Human Baseline)

Requirements:

- Create low-fidelity wireframes for TWO screens (Welcome/Home + Product Search Results)
- Consider elderly users (65+) and low digital literacy
- Include accessibility features
- Annotate key design decisions
- Provide 200-250 word design justification

SAGE Note: Students must design *FIRST* before AI critique. Shows their independent design thinking. Creates investment in their work—more likely to defend it thoughtfully against AI feedback rather than accept passively.

Part B: AI Heuristic Evaluation

Standardised Prompt:

"Evaluate these wireframes for a supermarket kiosk interface using Nielsen's 10 usability heuristics:

[Insert your wireframe description]

Target users: elderly customers (65+), customers with low digital literacy
Context: busy supermarket environment, standing interaction, 2-3 minute average session
Key tasks: product search, express ordering, viewing promotions

- For each identified issue:
- 1. State the specific heuristic violated
 - 2. Rate severity (1=cosmetic, 2=minor, 3=major, 4=catastrophic)
 - 3. Provide specific improvement recommendation
 - 4. Consider impact on elderly users specifically"

SAGE Note: Prompt provides context AI cannot know (elderly users, standing interaction, time pressure). Tests whether students catch context-inappropriate AI advice—e.g., AI suggesting features requiring dexterity or sustained attention.

Part C: Critical Response to AI Feedback

Students evaluate each AI suggestion using this matrix (minimum 10 items):

ID	AI FEEDBACK SUMMARY	HEURISTIC	AI SEVERITY	YOUR AGREEMENT	ACTION	IMPACT ON ELDERLY	JUSTIFICATION (50-75 WORDS)
F001	Increase font size to 24pt minimum	H8: Aesthetic	3 (major)	Agree	Implement	High	Vision decline is common in elderly users. Current 18pt too small for standing distance (~60cm from screen). 24pt aligns with WCAG AAA and elderly-specific HCI research. (68 words)
F002	Add live chat support button	H10: Help	2 (minor)	Partial	Modify	Medium	AI assumes staffed support, but GreenHarvest kiosks operate autonomously. Instead, add "Call Store Staff" button that lights help beacon at nearest checkout—simpler, matches user mental model. (72 words)

ID	AI FEEDBACK SUMMARY	HEURISTIC	AI SEVERITY	YOUR AGREEMENT	ACTION	IMPACT ON ELDERLY	JUSTIFICATION (50-75 WORDS)
F003	Implement gesture- based navigation (swipe)	H7: Flexibility	1 (cosmetic)	Disagree	Reject	High (negative)	Swipe gestures require fine motor control and prior smartphone experience. Elderly users (especially with arthritis) struggle with gestures. Large button taps are more accessible and familiar. AI ignored user capability constraints. (65 words)

✓ Required Distribution (Forces Discrimination)

- At least 4 items: Implement (accept AI suggestion)
- At least 3 items: Modify (AI's idea needs context adjustment)
- At least 3 items: Reject (AI advice inappropriate for this setting)

SAGE Note: The 4-3-3 distribution is deliberate. Research shows 90% of students confidently reject context-inappropriate AI advice when properly scaffolded. This quota ensures they practice that skill, building confidence to override AI in professional practice.

Part D: Comparative Analysis (200-250 words)

Required questions:

1. What valid usability issues did you identify that AI missed?
2. Which AI suggestions showed lack of context understanding?
3. How did AI feedback change your design perspective?
4. What makes human evaluation irreplaceable in this context?

SAGE Note: Question 4 is critical—forces students to articulate limits of AI automation. Builds appropriate skepticism. This is the payoff from the experiential learning approach: students discovered AI's blind spots firsthand, so their answer is grounded in evidence, not speculation.

4.3 Adapting SAGE to Your Discipline

The assessment patterns remain constant; only the content changes. Use this table to translate the examples above into your field:

Pattern 1: Requirements Synthesis Adaptation

DISCIPLINE	HUMAN BASELINE TASK	AI GENERATION PROMPT FOCUS	STANDARDS FOR EVALUATION	EXAMPLE ACCEPTANCE	EXAMPLE REJECTION
Nursing	8 care plan components for post-surgical patient	"Generate comprehensive discharge care plan considering mobility limitations, pain management, wound care, Indigenous cultural protocols"	NDIS Standards, clinical guidelines, cultural safety framework	Accept: AI's daily wound check schedule	Reject: AI's generic pain medication (contraindicated by patient's kidney function)
Business	8 communication strategies for change management	"Generate stakeholder communication plan for organizational restructure considering workforce diversity, remote workers, union presence"	Kotter's change model, organizational behavior research, employment law	Accept: AI's town hall meeting suggestion	Reject: AI's "all-staff email" ignores union consultation requirements under Fair Work Act
Engineering	8 safety requirements for construction site	"Generate safety management requirements for bridge construction considering height work, heavy machinery, public proximity, Indigenous heritage areas"	AS 1170, Safe Work Australia codes, environmental impact regulations	Accept: AI's fall protection harness requirement	Reject: AI omitted mandatory Indigenous heritage survey before excavation (state law requirement)

DISCIPLINE	HUMAN BASELINE TASK	AI GENERATION PROMPT FOCUS	STANDARDS FOR EVALUATION	EXAMPLE ACCEPTANCE	EXAMPLE REJECTION
Education	8 lesson plan elements for diverse classroom	"Generate inclusive lesson plan for Year 7 science considering EAL/D students, ADHD accommodations, gifted extensions, cultural diversity"	Australian Curriculum, UDL guidelines, state differentiation policy	Accept: AI's visual + text + video multimodal approach	Reject: AI's group work structure doesn't account for EAL/D language processing time

Pattern 2: Translation & Correction Adaptation

DISCIPLINE	STRUCTURED DESCRIPTION	AI TRANSLATION TARGET	COMMON AI ERRORS	PRE-TRANSLATION SCAFFOLD
Nursing	Clinical pathway description (8-10 steps)	Nursing process diagram or care pathway flowchart	Missing assessment data, incomplete SBAR, omitted delegation protocols	Assessment checklist; delegation matrix; SBAR template
Business	Decision-making process description	Business process model (BPMN)	Missing approval gates, incomplete risk assessment, unclear escalation paths	Approval authority matrix; risk register; escalation tree
Engineering	Construction sequence description	Critical path diagram or workflow chart	Missing safety holds, unrealistic durations, incomplete dependencies	Safety hold checklist; resource list; dependency matrix
Law	Legal procedure description (civil case)	Litigation process flowchart	Missing filing deadlines, incorrect jurisdiction, omitted discovery steps	Jurisdiction checklist; statutory deadline table; court rules summary

Pattern 3: Design Evaluation Adaptation

DISCIPLINE	DESIGN BASELINE	AI EVALUATOR ROLE	CONTEXT STUDENTS MUST PROVIDE	EXAMPLE CONTEXT-MISMATCH REJECTION
Nursing	Patient education handout wireframe	"Evaluate as health literacy consultant using Plain Language principles"	Patient demographic (age, education, primary language), diagnosis complexity, reading level	Reject: AI suggested medical terminology appropriate for healthcare workers, not patients with Year 8 reading level
Business	Organizational restructure proposal	"Evaluate as change management	Company culture, union presence, prior change fatigue,	Reject: AI recommended rapid rollout ignoring recent restructure 6

DISCIPLINE	DESIGN BASELINE	AI EVALUATOR ROLE	CONTEXT STUDENTS MUST PROVIDE	EXAMPLE CONTEXT-MISMATCH REJECTION
		consultant using Kotter's 8 steps"	workforce demographics	months ago (change fatigue high)
Education	Differentiated lesson plan	"Evaluate as instructional designer using UDL checkpoints"	Student cohort diversity, available technology, classroom layout, lesson duration	Reject: AI suggested iPad activities but school has 1:4 device ratio, not 1:1
Architecture	Accessible building design sketches	"Evaluate as accessibility consultant using Universal Design principles"	Building code jurisdiction, user demographics, budget constraints, climate	Reject: AI recommended features exceeding AS 1428 minimum requirements but over budget constraints

Universal Adaptation Checklist

For Any Discipline, Ensure Your SAGE Assessment Includes:

- ☐ **Human baseline first** — What students produce WITHOUT AI
- ☐ **Standardised prompt** — Ensures fairness; include context AI cannot know
- ☐ **Side-by-side submission** — Raw AI output + refined version
- ☐ **Decision log** — Accept/Modify/Reject with 50-75 word justifications
- ☐ **Standards/frameworks** — Domain-specific comparison anchors (regulations, clinical guidelines, professional codes)
- ☐ **Mandatory rejection quota** — Prevents passive acceptance (may become too difficult for non complex tasks hence the need for anchors)
- ☐ **Confidence ratings** — 1-5 scale for metacognitive awareness
- ☐ **Reflection component** — reflection (not generic but based on anchors) on AI's strengths/limitations in this context

4.4 What Makes These Assessments Work

1

Mandatory Human Baseline

Why it works: Prevents complete AI delegation; creates comparison anchor; shows student's independent capability; provides evidence of learning (not just AI output quality).

Pitfall if missing: Students submit polished AI output with shallow critique. You can't distinguish their knowledge from AI's.

2

Standardised Prompts

Why it works: Ensures fairness (all students get similar AI quality); includes context AI cannot know (elderly users, time pressure); surfaces whether students catch context-inappropriate advice.

Pitfall if missing: Prompt engineering skill becomes confounding variable. Unequal AI outputs make comparison unfair.

3

Mandated Quotas

Why it works: Forces discrimination (can't accept everything); builds confidence to override AI; prevents Level 1 (Passive Acceptor) behavior; ensures practice of all orchestration modes.

Pitfall if missing: Students accept 95% of AI suggestions with generic "looks good" justifications. No critical thinking demonstrated.

4

Evidence-Based Justifications

Why it works: 50-75 word requirement forces specific reasoning; citations to research which prevent generic statements; research shows $p=0.84$ correlation between justification quality and evidence use.

Pitfall if missing: "AI was incomplete" or "AI missed details" without specifics. No learning demonstrated.

5

Pre-Translation Scaffolds

Why it works: Addresses predictable AI failure modes (boundaries 55%, exceptions 55%); forces attention to critical elements BEFORE AI; makes errors more visible during correction.

Pitfall if missing: Students and AI both miss the same things. No learning trigger occurs.

6

Reflection Requirements

Why it works: Forces metacognition; builds awareness of AI's pattern of strengths/weaknesses; develops appropriate skepticism; transfers to future professional practice beyond this assessment.

Pitfall if missing: Students complete task mechanically without internalizing lessons about AI limitations.

Common Implementation Mistakes and Fixes

MISTAKE	WHY IT FAILS	SAGE FIX
"Use AI to help with this assignment"	Vague permission leads to undisclosed use; can't distinguish student from AI work; no learning	Mandatory side-by-side submission; exact prompts documented; decision logs required
Rubric awards marks for "output quality"	AI can produce polished prose; marks go to AI, not student thinking	70% of marks for process (justification, gap detection, standards mapping); 30% for output
No mandatory rejection requirement	Students accept 95% of AI; minimal critical engagement; passive acceptance becomes norm	Require rejection of $\geq 20\%$ of AI suggestions with contextual justification
Generic feedback: "AI was wrong"	No evidence of domain knowledge; could be guessing; doesn't demonstrate learning	50-75 word justifications with specific citations to standards/research/stakeholder evidence
Students use different AI tools	Quality varies dramatically (GPT-4 vs GPT-3.5); unfair comparison; confounds results	Standardized prompts partially mitigate; OR specify one tool; OR provide AI output to all students
No human baseline required	Can't prove student understands task independently; pure AI delegation possible	Always require human work FIRST (bullet points, hand sketch, interview notes)
Reflection is optional or generic	No metacognition; lessons about AI limitations not internalized; no transfer to future work	Mandatory 200-400 word reflection with specific questions targeting competency dimensions

Rubric Design: Mark the Process, Not the Product

Recommended Weighting Distribution		
CRITERION	WEIGHT	WHAT YOU'RE ACTUALLY ASSESSING
Justification Quality & Evidence	25-30%	Specific citations to standards/research; reasoning depth; evidence-based modifications
Gap/Error Detection	20-25%	Finding AI mistakes using error taxonomy; identifying missing components; boundary reasoning
Contextual Reasoning	15-20%	When and why to override AI; stakeholder considerations; environmental constraints
Standards Mapping	15-20%	Correct alignment with regulations/frameworks; demonstrating domain expertise

CRITERION	WEIGHT	WHAT YOU'RE ACTUALLY ASSESSING
Metacognitive Reflection	10-15%	Self-awareness about AI strengths/limitations; articulating when AI fails; transfer insights
Output Quality	10-15%	Professional presentation; completeness (AI can help here legitimately)

Key principle: 70-85% of marks assess thinking and reasoning (the process); only 10-15% assess polish (the product AI can produce). This inverts traditional rubrics.

When Students Struggle: Diagnostic Indicators

Level 1 (Passive Acceptor) Red Flags

- Accepts >90% of AI suggestions
- Justifications generic: "AI was good" or "looks complete"
- No specific citations to standards
- Reflection focuses on AI convenience, not limitations
- Confidence ratings all 4-5/5

Intervention: Provide worked example showing L3 justifications; schedule viva to probe reasoning; assign additional tutorial with explicit scaffolding

Level 2 (Selective Adapter) Yellow Flags

- Accepts ~70%; modifies ~25%; rejects ~5%
- Justifications cite standards but lack depth
- Struggles with boundary reasoning
- Misses exceptions and edge cases
- Reflection shows awareness but limited articulation

Intervention: Provide boundary identification exercises; emphasize CRUD completeness; model deep justification; target likely to reach L3 with practice

Success Indicators: You're Doing SAGE Right When...

✓ Green Lights

- Students **confidently reject** AI advice with contextual reasoning (90% do this in design evaluation when properly scaffolded)
- Justifications cite **specific standards** (NIST control A.6.2.1, Privacy Act APP 11, NDIS Standard 3.2) not generic principles
- Reflections articulate **AI's pattern of failures** ("AI consistently misses enforcement mechanisms and third-party vendor risks")
- Students use **domain expertise** to override AI ("AI suggested X but our elderly user context requires Y instead")
- Modification ratios **balance** across Accept/Modify/Reject (not 95/5/0)
- Students report **time saved on drafting, time gained for analysis** (the ideal outcome)
- Orchestration distribution: **70%+ at Level 3** (Balanced Integrator) by end of semester
- Students learn to spot AI hallucinations, bias and are security aware

Final Implementation Checklist

Before Deploying Your SAGE Assessment:

- ☐ Reviewed Section 3 framework and can explain the 4-step cycle to students
- ☐ Selected appropriate assessment pattern (synthesis, translation, or evaluation)
- ☐ Developed human baseline task students complete FIRST
- ☐ Created standardised prompt including context AI cannot know
- ☐ Designed decision log template with justification requirements (50-75 words)
- ☐ Included mandatory rejection quota ($\geq 20\%$ of AI suggestions)
- ☐ Adapted rubric to weight process (70%) over product (30%)
- ☐ Prepared domain-specific standards/frameworks for comparison
- ☐ Written reflection questions targeting all three competency dimensions
- ☐ Created worked example showing L1 vs L3 justification quality
- ☐ Set up monitoring indicators (orchestration levels, accessibility articulation, modification rates)
- ☐ Planned viva questions for spot-checking authorship if needed

You now have everything needed to implement SAGE next term.

The next sections provide templates, FAQs, and troubleshooting guidance to support your implementation journey.

SECTION 5

About This Resource and Further Support

This section provides comprehensive information about the authors, detailed research evidence, references, licensing, companion resources, and implementation support to assist your SAGE adoption journey.

5.1 Who We Are

Dr Mahmoud Elkhodr

Senior Lecturer
School of Engineering and Technology
Central Queensland University, Australia

✉ **Email:** m.elkhodr@cqu.edu.au

Dr Elkhodr developed the SAGE framework through systematic empirical research involving 500+ students across six published studies. His work focuses on translating AI pedagogy research into practical, implementable assessment designs for higher education.

Research specialisation:

Cybersecurity education, AI-enhanced pedagogy, responsible GenAI integration, IoT security, assessment design

Professor Ergun Gide

Professor of Information Systems
School of Engineering and Technology
Central Queensland University, Australia

✉ **Email:** e.gide1@cqu.edu.au

Professor Gide co-developed the SAGE framework and led multi-institutional validation research across business, ICT, and systems analysis disciplines. His research explores how emerging technologies can be integrated into educational practice whilst maintaining academic integrity.

Research specialisation:

Information systems education, enterprise systems, technology adoption in higher education, GenAI applications, digital transformation

Central Queensland University's Leadership in GenAI Education

The SAGE framework represents one of Australia's—and potentially the world's—pioneering efforts to systematically integrate generative AI into formal higher education curricula with empirical validation. Central Queensland University has supported this research program through:

- Multi-campus implementation enabling cross-institutional validation
- Support for assessment-as-research methodologies that simultaneously teach and generate evidence
- Commitment to open scholarship through Creative Commons licensing of educational resources
- Investment in faculty development and pedagogical innovation in AI-enhanced teaching

Collaborative Research Program

The SAGE framework emerged from a multi-year collaborative research program between Drs Elkhodr and Gide exploring responsible AI integration across diverse disciplines. This work builds on:

- **30+ published articles** on GenAI in education spanning empirical studies, theoretical frameworks, and domain-specific implementations
- **Cross-disciplinary validation** in cybersecurity, data analytics, systems analysis, business information systems, and ERP education
- **Industry engagement** ensuring alignment with professional practice and employer expectations

This guide represents the distillation of that research into immediately actionable guidance for educators worldwide.

5.2 Complete Research Evidence

Purpose of this section: Provide comprehensive methodology and detailed findings for educators seeking institutional approval, conducting research replication, or validating SAGE in their own contexts. If you're primarily interested in implementation, Sections 1-4 contain everything you need.

Study 1: ICT Students and ChatGPT Performance (Experimental Labs)

Research Design

Sample: 52 students across three cohorts (15 UG on-campus, 18 UG online, 19 PG)

Method: Controlled experimental design with two tasks:

- Task 1 (Baseline):** System design using search engines only—no AI permitted
- Task 2 (Intervention):** Similar system design using ChatGPT with structured guidance

Assessment: Rubric-based evaluation across four dimensions—Functionality & User Flow, Content & Information Hierarchy, Visual Design, Technical Implementation (each scored 0-4)

Data collection: Performance scores, post-task reflections, observation notes

Quantitative Results

Metric	Cohort 1 (UG, N=15)	Cohort 2 (UG Online, N=18)	Cohort 3 (PG, N=19)	Average Gain
Functionality & User Flow	1.6 → 3.0 (+87.5%)	1.4 → 3.0 (+85.6%)	1.8 → 3.2 (+77.8%)	+83.6%
Content & Information Hierarchy	2.4 → 3.0 (+25%)	1.8 → 3.0 (+28%)	2.35 → 3.15 (+34%)	+29.0%
Visual Design	2.0 → 2.8 (+40%)	1.9 → 2.9 (+53%)	2.1 → 2.9 (+38%)	+43.7%
Overall Performance	+56.3%	+56.8%	+55.9%	+56.3%

Qualitative Findings

Student perceptions (post-task reflections, n=52):

Perception	Cohort 1	Cohort 2	Cohort 3	Pattern Observed
"Very Helpful/Helpful"	80.0% (12/15)	66.7% (12/18)	36.8% (7/19)	PG students less enthusiastic than UG

PERCEPTION	COHORT 1	COHORT 2	COHORT 3	PATTERN OBSERVED
"Very Enjoyable/Enjoyable"	86.7% (13/15)	83.3% (15/18)	52.7% (10/19)	UG more positive about experience
"Likely to use AI in future"	40.0%	44.4%	26.3%	Cautious adoption across all groups

Key Insight

Interpretation: Postgraduate students found ChatGPT less helpful than undergraduates despite similar performance gains. Possible explanations: (1) PG students have established workflows AI disrupts, (2) PG cohort included international students less familiar with conversational AI, (3) PG tasks were more complex, revealing AI limitations more clearly. This informed SAGE's differentiated scaffolding for UG vs PG contexts.

Study 2: Data Analytics Students (Survey + Workshop Implementation)

Research Design

Sample: 74 students (73% UG, 27% PG; 65% business background, 35% ICT)

Method: Mixed-methods design:

- Video introduction to ChatGPT capabilities
- Hands-on tutorial with data analytics use cases
- 27-item survey (5-point Likert scale) measuring perceived benefits, challenges, satisfaction, and engagement

Analysis: Chi-square tests for association; effect size calculations (η^2); thematic coding of open-ended responses

Statistical Results

- **Perceived Benefits:** $\eta^2=0.173$ (large effect) — ChatGPT accounts for 17.3% of variance in improved academic outcomes
- **Perceived Challenges:** $\eta^2=0.289$ (large effect) — Significant concerns about complex query handling (35% of students), lack of human interaction (30%), and privacy (25%)
- **Overall Satisfaction:** Mean = 3.88/5 (SD=0.92)
- **Impact on Academic Engagement:** Mean = 3.18/5 (SD=1.05)
- **Frequency-Benefit Correlation:** Chi-square significant ($p<.05$) — More frequent users perceive greater benefits

Critical Finding

Challenges accounted for **more variance (28.9%) than benefits (17.3%)**, suggesting that without structured scaffolding (which this study lacked), students' negative experiences outweigh positive ones. This finding directly motivated the development of SAGE's explicit guidance structures in subsequent studies.

Study 3: Cybersecurity Management (Full SAGE Implementation)

Research Design

Sample: 105 students (72 UG, 33 PG) over 12-week semester at CQU Sydney campus

Method: Two-stage SAGE implementation with three assessments:

- **Week 4:** Email policy development (tutorial-based with high scaffolding)
- **Week 8:** Risk assessment using NIST CSF (moderate scaffolding)
- **Week 12:** Incident response plan with industry engagement (low scaffolding)

Data collection: Assignment submissions (side-by-side artefacts), reflection essays, instructor observations, end-of-semester survey

Analysis: Qualitative rubric-based coding; thematic analysis of reflections; descriptive statistics on completion and self-reported outcomes

Quantitative Outcomes

- **Assignment completion rates:** Exceeded 90% across both UG and PG cohorts
- **Self-reported critical thinking improvement:** 78% of students (82/105)
- **Value of real-world assessment tasks:** 87% agreed or strongly agreed (91/105)
- **Initial struggle with gap identification:** 45% (47/105) struggled to identify contextual gaps in AI outputs in Week 4
- **Improvement by Week 12:** Only 12% (13/105) still struggled with gap identification

Qualitative Themes from Student Reflections

Representative Student Quotes:

"The AI-generated policy was a great starting point, but refining it based on research allowed me to understand the gaps and create a more tailored solution." — UG student, Week 8 reflection

"Using GenAI helped me focus on improving content rather than starting from scratch, saving time and allowing me to dive deeper into compliance and security details." — PG student, Week 12 reflection

"The AI-generated policy was a good outline, but it lacked enforcement mechanisms and did not include risk mitigation plans for third-party vendors. AI alone is insufficient for developing robust cybersecurity policies." — PG student, final reflection

Instructor Observations (Systematic Notes)

- Notable improvement in students' ability to assess regulatory compliance and adapt security policies to organisational needs
- Students who engaged with industry partners (where available) demonstrated stronger contextual reasoning than those using case studies alone
- Over-reliance on AI-generated outputs decreased progressively from Week 4 → Week 12 as scaffolding reduced
- Variability in students' familiarity with AI tools required targeted scaffolding in early weeks

Study 4: Systems Analysis & Design (Multi-Campus SAGE Validation)

Research Design

Sample: 18 student groups (~60-70 students total) across 4 campuses (Brisbane baseline, Melbourne, Sydney, Online)

Method: Three embedded experiments functioning simultaneously as pedagogy, assessment, and research:

- **Experiment 1:** Requirements synthesis (Accept/Modify/Reject with justifications and confidence ratings)
- **Experiment 2:** DFD translation and correction (error taxonomy and correction logs)
- **Experiment 3:** Design evaluation (4-3-3 Implement/Modify/Reject distribution with contextual justifications)

Analysis: Descriptive statistics on orchestration distribution; Spearman rank correlations among rubric-coded measures; thematic coding of justifications; error classification

Orchestration Level Distribution

LEVEL	PERCENTAGE	CHARACTERISTICS	DECISION PATTERN
Level 4: Critical Synthesiser	0%	Proactive gap-finding; novel solutions beyond AI	N/A — None achieved
Level 3: Balanced Integrator	46%	Evidence-based justifications; boundary reasoning	~50% accept, ~40% modify, ~10% reject
Level 2: Selective Adapter	38%	Surface reasoning; struggles with exceptions	~70% accept, ~25% modify, ~5% reject
Level 1: Passive Acceptor	15%	Minimal critique; justifications generic	>90% accept, <10% modify/reject



Interpretation

The absence of Level 4 appears to be a task design issue rather than student capability ceiling. All three experiments were framed as evaluation/correction tasks. A subsequent informal trial with one "gap-finding" assessment (deliberately under-specified requirement) produced 2 groups reaching Level 4. **Implication:** Include at least one adversarial/synthesis task to develop proactive orchestration.

Competency Correlations (Spearman Rank ρ)

COMPETENCY PAIR	P (SPEARMAN)	INTERPRETATION
Justification Quality ↔ Synthesis Evidence	0.84	Students who cite evidence write significantly better justifications
Synthesis Evidence ↔ NFR Breadth	0.78	More non-functional requirements = better synthesis capability
Domain Knowledge ↔ Accessibility Articulation	≈1.00	Technical depth strongly predicts accessibility awareness
Context Calibration ↔ Readiness to Reject AI	0.85	Contextual reasoning practice builds confidence to override AI

Implication: AI orchestration develops as a competency cluster. Teaching one dimension (e.g., evidence-based justification) improves related dimensions (synthesis, NFR awareness, rejection confidence). Rubrics should assess multiple dimensions, not just correctness.

AI-to-DFD Error Distribution (n=104 errors detected)

ERROR CATEGORY	PERCENTAGE	EXAMPLE
Structural	30%	Incorrect flow directions; processes feeding back to themselves
Completeness	27%	Missing CRUD operations; omitted data stores
Semantic	26%	Misunderstood business logic; conflated entities
Notation	15%	Incorrect symbols; violated DFD conventions
Scope	2%	Minor boundary issues

Cross-cutting struggles (% of groups):

- **Boundary reasoning:** 55% struggled to define system scope correctly
- **State/CRUD management:** 45% missed state-change operations
- **Exception handling:** 55% omitted error pathways

The Accessibility U-Curve (Validated Pattern)

TASK/ABSTRACTION LAYER	ACCESSIBILITY ARTICULATION (%)	WHAT STUDENTS ADDRESSED
Requirements (prose)	85%	Mentioned elderly users, assistive technology, inclusive design needs
Architecture (DFD/ERD)	10%	Rarely included accessibility in diagrams or architectural decisions
Interface (wireframes)	90%	Large fonts, clear labels, simplified navigation for elderly users

Mechanisms identified:

1. **Cognitive salience:** Access needs feel relevant to interfaces, "distant" from architecture
2. **Notational affordance:** DFD/ERD lack obvious places to encode capability profiles
3. **Cognitive load:** Complex diagrams crowd out secondary concerns

Intervention result: When micro-notation (AccessibilityProfile store, AssistanceOrchestrator process) and two mandatory prompts were added in subsequent iteration, middle-phase articulation increased from 10% to 76%.

Study 5: AI as Critic (SAGE Pedagogy Validation for Human Authority)

Research Design

Focus: Validation of SAGE pedagogy for maintaining human authority and responsible GenAI use in Systems Analysis and Design education

Method: Students used AI as a critical evaluator of their own work, reversing the typical AI-as-generator pattern. This "AI as Critic" approach tested whether students could maintain intellectual authority when AI provided feedback rather than content.

Key Finding: The critic-role framing strengthened students' sense of ownership over their work while still benefiting from AI feedback, supporting the SAGE principle that humans must remain the ultimate decision-makers.

Publication: EdArXiv Preprints, 2025. Available: <https://osf.io/preprints/edarxiv/8j3xf>

Study 6: Empirical Typologies of Student GenAI Use in Permitted Assessments

Research Design

Sample: 167 first-year ICT students (behavioral data) and 163 reflective responses

Method: Mixed-methods study examining actual ChatGPT usage during a supervised in-class assessment under institutional AI Collaborate permissions. Data collected through embedded 12-item reflection instrument capturing behavioral patterns and qualitative reflections.

Key Findings: Identified four distinct student typologies (Strategic Optimisers 32%, Dialogic Learners 28%, Cautious Adopters 23%, Experimental Users 17%) and discovered a "Goldilocks Zone" of 4-8 prompts for optimal engagement. Revealed competency-confidence inversion wherein students demonstrate

sophisticated AI interaction strategies whilst experiencing regulatory anxiety. 77.8% struggled with verification competencies despite 73% demonstrating systematic verification behaviors. AI functioned as linguistic equalizer for international students (46.7% citing English confidence).

Publication: Submitted to *Discover Education*, 2026. Preprint: <https://doi.org/10.21203/rs.3.rs-8628653/v1>

Limitations Across All Studies

Acknowledged Constraints

- **Sample size:** Modest (n=52-105 per study); limits statistical generalisability
- **Geographic scope:** Australian universities only; cultural transferability unknown
- **Discipline coverage:** ICT-heavy (cybersecurity, systems analysis, data analytics); broader validation needed
- **Single-institution bias:** Most studies at Central Queensland University; institutional culture effects possible
- **Self-report reliance:** Perceptions ≠ objective competency; need performance metrics
- **Tool specificity:** ChatGPT-focused (GPT-3.5/4 era); other tools may behave differently
- **Short timeframes:** Single-semester observations; no longitudinal retention data
- **Lack of control groups:** Ethical constraints prevented withholding SAGE; quasi-experimental designs used
- **Coder reliability:** Rubric application consistency not yet fully reported across all studies

Despite these limitations: Patterns replicated across 6 studies, 500+ students, 3 disciplines, and 4 campuses suggest robustness. The framework's theoretical grounding in constructivism, Bloom's taxonomy, and situated learning provides additional validity beyond empirical evidence alone.

5.3 References

Core SAGE Framework Publications

The following six peer-reviewed publications form the empirical foundation of the SAGE framework:

[1] M. Elkhodr, E. Gide, R. Wu, and O. Darwish, "ICT students' perceptions towards ChatGPT: An experimental reflective lab analysis," *STEM Education*, vol. 3, no. 2, pp. 70-88, 2023. [Online]. Available: <https://doi.org/10.3934/steme.2023006>

[2] R. Sandu, E. Gide, and M. Elkhodr, "The role and impact of ChatGPT in educational practices: Insights from an Australian higher education case study," *Discover Education*, vol. 3, no. 1, art. 71, 2024. [Online]. Available: <https://doi.org/10.1007/s44217-024-00126-6>

[3] M. Elkhodr and E. Gide, "Integrating Generative AI in Cybersecurity Education: Case Study Insights on Pedagogical Strategies, Critical Thinking, and Responsible AI Use," *Discover Education*, vol. 4, art. 225, Nov. 2025. doi: 10.1007/s44217-025-00935-3

[4] M. Elkhodr and E. Gide, "Embedding Generative AI into Systems Analysis and Design Curriculum: Framework, Case Study, and Cross-Campus Empirical Evidence," *arXiv preprint arXiv:2511.17515*, Nov. 2025. doi: 10.48550/arXiv.2511.17515

[5] M. Elkhodr and E. Gide, "AI as Critic: Validating SAGE Pedagogy for Human Authority and Responsible GenAI Use in Systems Analysis and Design Education," *EdArXiv Preprints*, 2025. [Online]. Available: <https://osf.io/preprints/edarxiv/8j3xf>

[6] M. Elkhodr, A. Azra, and E. Gide, "How First-Year Students Actually Use ChatGPT in Permitted Assessments: Empirical Typologies, Verification Gaps, and the Policy-Practice Divide," *submitted to Discover Education*, 2026. Preprint available: <https://doi.org/10.21203/rs.3.rs-8628653/v1>

Extended Research Portfolio: GenAI in Education

The SAGE framework builds on a comprehensive research program exploring generative AI integration across diverse educational contexts. Selected works addressing complementary aspects of AI-enhanced pedagogy include:

Foundational Works

[7] M. Elkhodr and E. Gide, Eds., *Generative Artificial Intelligence Empowered Learning: A New Frontier in Educational Technology*, 1st ed. New York, NY: Chapman and Hall/CRC, 2025, 248 pp. doi: 10.1201/9781003422433. eBook ISBN: 9781003422433

Cross-Sectoral and Systematic Reviews

- R. J. Eddine, E. Gide, and A. Al-Sabbagh, "Generative AI in higher education: A cross-sector analysis of ChatGPT's impact on STEM, social sciences, and healthcare," *STEM Education*, vol. 5, no. 5, pp. 757–801, 2025. doi: 10.3934/steme.2025035
- K. Wangsa, S. Karim, E. Gide, and M. Elkhodr, "A systematic review and comprehensive analysis of pioneering AI chatbot models from education to healthcare: ChatGPT, Bard, Llama, Ernie and Grok," *Future Internet*, vol. 16, no. 7, art. 219, 2024. doi: 10.3390/fi16070219
- K. Wangsa, R. Sandu, S. Karim, M. Elkhodr, and E. Gide, "A systematic review and analysis on the potentials and challenges of GenAI chatbots in higher education," in *Proc. 21st Int. Conf. Information Technology Based Higher Education and Training (ITHET)*, Paris, France, Nov. 2024, pp. 1–7. doi: 10.1109/ITHET61869.2024.10837608
- A. Al Tawara, J. El-Den, E. Gide, and Y. Sebastian, "A systematic review and comprehensive analysis of AI-enabled re-skilling and upskilling in education: Transformative strategies for the future," in *Proc. 21st Int. Conf. Information Technology Based Higher Education and Training (ITHET)*, Paris, France, Nov. 2024, pp. 1–10. doi: 10.1109/ITHET61869.2024.10837638
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Discipline-Specific GenAI Integration

- H. Ranasinghe, E. Gide, and M. Elkhodr, "The significance of GenAI empowered ERP systems course teaching in quality education," in *Proc. 21st Int. Conf. Information Technology Based Higher Education and Training (ITHET)*, Paris, France, Nov. 2024, pp. 1–7. doi: 10.1109/ITHET61869.2024.10837679
- G. Chaudhry, E. Gide, E. Yadegaridehkordi, and R. Tumpa, "Generative AI-powered teaching and learning in engineering and project management higher education: A systematic review," in *Joint International Conference on AI, Big Data and Blockchain*, Cham, Switzerland: Springer Nature, Aug. 2025, pp. 99–113.
- E. Gusman, E. Gide, M. Elkhodr, and G. Chaudhry, "The benefits and challenges of using artificial intelligence in teaching English as a foreign language in higher education," in *Proc. 21st Int. Conf. Information Technology Based Higher Education and Training (ITHET)*, Paris, France, Nov. 2024, pp. 1–7. doi: 10.1109/ITHET61869.2024.10837597
- E. Gusman, E. Gide, G. Chaudhry, and M. Elkhodr, "A comprehensive review to identify the challenges and opportunities of using digital technology in English teaching in higher education," in *International Society for Technology, Education, and Science*, 2023.

Curriculum and Framework Development

- R. Sandu, E. Gide, S. Karim, and P. Singh, "A framework for GenAI-empowered curriculum and learning resources: A case study from an Australian higher education," in *Proc. 21st Int.*

Conf. Information Technology Based Higher Education and Training (ITHET), Paris, France, Nov. 2024, pp. 1–8. doi: 10.1109/ITHET61869.2024.10837623

- N. Sandu and E. Gide, "Adoption of AI-Chatbots to enhance student learning experience in higher education in India," in *Proc. 18th Int. Conf. Information Technology Based Higher Education and Training (ITHET)*, Sept. 2019, pp. 1–5. doi: 10.1109/ITHET46829.2019.8937382

Equity, Ethics, and Interdisciplinary Integration

- M. Elkhodr, K. Wangsa, E. Gide, and S. Karim, "A systematic review and multifaceted analysis of the integration of artificial intelligence and blockchain: Shaping the future of Australian higher education," *Future Internet*, vol. 16, no. 10, art. 378, 2024. doi: 10.3390/fi16100378
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- N. Abbasi, E. Gide, P. Kalutara, and P. Lawrence, "Barriers and opportunities in online delivery of architecture and building design studios: Australian educators' perspectives," 2024.

Note: For research collaborations or institutional partnerships, contact details are provided in Section 5.6.

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5.4 How to Use and Cite This Guide

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When referencing this guide in academic work, institutional documents, or educational materials, please use the appropriate citation format:

IEEE Style

M. Elkhodr and E. Gide, "Embedding Generative AI in Curriculum: The SAGE Framework and Evidence-Based Implementation Guide," Central Queensland University, Rockhampton, Australia, Version 1.0, Jan. 2026. [Online]. Available: <https://doi.org/10.5281/zenodo.18383952>

APA Style (7th Edition)

Elkhodr, M., & Gide, E. (2026). *Embedding generative AI in curriculum: The SAGE framework and evidence-based implementation guide* (Version 1.0). Central Queensland University. <https://doi.org/10.5281/zenodo.18383952>

Harvard (Australian) Style

Elkhodr, M and Gide, E 2026, *Embedding generative AI in curriculum: The SAGE framework and evidence-based implementation guide*, Version 1.0, Central Queensland University, Rockhampton, viewed [date], <<https://doi.org/10.5281/zenodo.18383952>>.

MLA Style (9th Edition)

Elkhodr, Mahmoud, and Ergun Gide. *Embedding Generative AI in Curriculum: The SAGE Framework and Evidence-Based Implementation Guide*. Version 1.0, Central Queensland University, Jan. 2026, <https://doi.org/10.5281/zenodo.18383952>.

Attribution for Adapted Materials

If you adapt assessment templates, rubrics, or other materials from this guide, please include:

"This [assessment/rubric/template] is adapted from the SAGE Framework by Elkhodr & Gide (Central Queensland University, 2026), licensed under CC BY 4.0. Original available at <https://doi.org/10.5281/zenodo.18383952>. Modifications include: [brief description of changes]."

Permissions

✓ You Do Not Need Permission To:

- Use SAGE in your courses (any discipline, any level)
- Adapt assessment templates to your context
- Share this guide with colleagues or students
- Translate this guide to other languages (with attribution)
- Include excerpts in institutional policy documents
- Use materials in faculty development workshops

Please Contact Us If You:

- Publish research using SAGE (we'd love to know!)
- Create substantial derivative works for publication
- Want to collaborate on multi-institutional validation
- Have suggestions for improving the framework

Contact details in Section 5.6

5.5 Companion Resources

This educator's guide is part of a comprehensive SAGE implementation ecosystem. Additional resources are available to support your adoption:

Student-Facing GenAI Literacy Module

 GenAI Literacy and Numeracy 101 for Higher Education Students

Format: Interactive HTML module + Static PDF version

Purpose: Equip your students with foundational GenAI competencies before they begin SAGE assessments. This module addresses the "variable AI literacy" challenge observed in our research.

Module Contents (10 Lessons)

1. Module Overview Introduction to GenAI literacy, learning outcomes, and structure	2. Lesson 1: What is Generative AI? Understanding foundations, capabilities, and limitations (LLMs)
3. Lesson 2: How to Talk to AI Mastering basic prompt engineering (Instruction & Role Prompting)	4. Lesson 3: Using AI the Right Way Exploring responsible use, ethics, and academic integrity
5. Lesson 4: Practical Applications Examples across disciplines: note-taking, explaining concepts, etc.	6. Lesson 5: Personal Tutor Using the "Feedback Loop" to critique and improve your work
7. Lesson 6: Finding Academic Sources Locating and verifying credible research; avoiding hallucinations	8. Lesson 7: Group Projects Fair and transparent AI use in team assignments
9. Final Quiz Scenario-based questions to test your knowledge	10. Module Wrap-Up Key takeaways, official resources, and next steps

 [Access the GenAI Literacy and Numeracy Module via Zenodo](#) | Completion time:
45-60 minutes
DOI: 10.5281/zenodo.18295619

Upcoming Resources (2026)

Additional implementation resources currently under development:

- **SAGE Implementation Toolkit:** Downloadable templates including assessment task descriptions, decision log templates, process-focused rubrics, standardised prompts for multiple disciplines, and monitoring spreadsheets (all licensed CC BY 4.0)
- **Worked exemplars:** Anonymised student work showing Level 1 vs Level 3 orchestration quality with annotations
- **Tutorial slides:** PowerPoint presentations for introducing SAGE to students (customisable)

These resources will be made available via institutional repository and linked in future versions of this guide.

5.6 Implementation Support

Contact the Authors

Dr Mahmoud Elkhodr

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Profile: <https://staff-profiles.cqu.edu.au/home/view/1633>

Professor Ergun Gide

Email: e.gide1@cqu.edu.au

Profile: <https://staff-profiles.cqu.edu.au/home/view/956>

What We Can Help With

- Adapting SAGE to your discipline-specific context
- Designing assessment tasks and rubrics for your unit
- Interpreting orchestration level distributions in your cohort
- Troubleshooting implementation challenges
- Establishing research partnerships for multi-institutional validation
- Presenting SAGE at your institution (faculty development workshops)

Institutional Partnerships

For universities seeking comprehensive support:

Faculty Development Programs

We offer customised professional development workshops (half-day or full-day) for academic staff implementing SAGE across multiple disciplines. Workshop includes: framework explanation, hands-on assessment design, rubric development, peer review calibration, and implementation planning. Contact us for availability and fees.

Research Collaboration

Interested in multi-institutional SAGE validation or conducting formal research on AI orchestration? We welcome partnerships for ethics-approved studies, joint publications, and cross-campus implementation research. PhD students investigating GenAI pedagogy are especially encouraged to reach out.

5.7 Version History and Updates

VERSION	RELEASE DATE	CHANGES
1.0	January 2026	Initial public release • Complete SAGE framework documentation • Evidence from 6 published studies • Implementation examples (formative + summative) • Cross-discipline adaptation guides • Companion student literacy module (DOI: 10.5281/zenodo.18295619)
1.1	Planned Q3 2026	<i>Anticipated updates based on community feedback:</i> • Implementation toolkit (templates, rubrics, worked exemplars) • Additional discipline-specific examples • Expanded troubleshooting section • Enhanced guidance for large cohorts

How to Stay Updated



Updates and Announcements

- Check the authors' institutional research profiles for new publications and resources
- Contact the authors directly (Section 5.6) to be notified of significant updates
- Monitor TEQSA GenAI Knowledge Hub for sector-wide developments

Contributing to SAGE Development

We welcome contributions from the educator community:

- **Case studies:** Share your implementation experiences (successes and challenges)
- **Discipline adaptations:** Contribute templates for underrepresented fields
- **Research data:** Share your orchestration distribution findings for meta-analysis
- **Student exemplars:** Anonymised work showing L1-L4 progression

Contributors will be acknowledged in future versions. Contact details in Section 5.6.

You're Ready to Implement SAGE

This guide has equipped you with the framework, evidence, examples, and templates to integrate generative AI responsibly into your curriculum starting next term.

Next Step: Equip Your Students

Before students begin SAGE assessments, assign the companion module:
"GenAI Literacy and Numeracy 101 for Higher Education Students"
(10 interactive lessons + quiz | 45-60 minutes)

Educator Guide

This document

Student Module

GenAI Literacy 101 (Zenodo)

Access student module: <https://doi.org/10.5281/zenodo.18295619>

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